

Metathesis, syllable weight and stress in Sevillian Spanish*

Madeline Gilbert, University of Georgia
madeline.gilbert@uga.edu

November 2024

Abstract

This study focuses on stress and metathesis in Sevillian Spanish, investigating how their interaction sheds light on phonological representations. Metathesis affects /s/-voiceless stop sequences, moving a debuccalized coda /s/ to the release of the following stop (/pasta/ → [patha]). This process plausibly changes syllable structure: the syllable where /s/ originated is closed at one representational level, but open on the surface ([pah.ta] → [pa.tha]). The change in syllable structure could affect weight-sensitive stress, depending on which form listeners reference for stress assignment. I present results from a stress judgment task in which Sevillian listeners treated syllables where /s/ had metathesized out similarly to heavy penults, and differently from light penults. I outline several analyses to account for their behavior, and suggest that a comprehensive analysis could (a) include gestural representations; (b) separate stress from metathesis, so that phonetic variability in the realization of metathesis is permitted but does not affect stress.

1 Introduction

In Sevillian Spanish, orthographic <s> + voiceless stop clusters undergo several changes, including metathesis. Coda /s/ reduces to [h] (debuccalization), producing *h-stop* sequences, and [h] metathesizes with the adjacent stop, producing *stop-h* sequences (1) (Torreira 2006; O'Neill 2010; Parrell 2012; Torreira 2012; Ruch and Harrington 2014; Ruch and Peters 2016; Henriksen, Galvano, and Fischer 2023).

*This work was developed at NYU: thanks to Maria Gouskova, Lisa Davidson, Gillian Gallagher, Juliet Stanton, Kate Mooney, and the NYU PEP Lab for their helpful feedback throughout its development. I'd also like to thank two anonymous *Phonology* reviewers and Eric Baković, as well as the Associate Editor and Editor Yoonjung Kang, for constructive input that contributed substantially to the development of this paper, especially to the theoretical portions.

This work was largely done at New York University. It was finished in Paris, France, during my time at the Laboratoire de Phonétique et Phonologie. There, this work was partially supported by a public grant, overseen by the IdEx Université Paris Cité (ANR-18-IDEX-0001) as part of the Labex Empirical Foundations of Linguistics - EFL.

(1) Sevillian h-stop and stop-h sequences

	Debuccalization	Metathesis	
a. /'tʃispa/	['tʃihpa]	['tʃipha]	‘spark’
b. /'pista/	['pihta]	['pitha]	‘track’
c. /'boske/	['bohke]	['bokhe]	‘forest’

Metathesis is an ongoing change in this variety and can occur in all /s/-voiceless stop sequences. It occurs in all stress configurations (Ruch 2008; Torreira 2012; Horn 2013) and both within morphemes and across word and morpheme boundaries (Ruch 2008; Horn 2013). The realization of <s> + voiceless stop sequences is highly variable, and metathesis is only one possible variant in Sevillian (Section 2.1).

Metathesis raises the following question: are the syllables /s/ metathesizes out of heavy or light? The answer sheds light on the representation of these sequences and on the structure of the phonological grammar. Metathesis in /sC/ sequences could produce a mismatch between segmental order and syllable structure at different representational levels or derivational stages. This mismatch permits us to disentangle surface forms from non-surface forms by designing experiments so that listeners’ responses can be explained by reference to one—but not another—level of representation. More broadly, the mismatch creates an opportunity to investigate the separation between components of the phonological grammar and the role of abstraction.

The possibility that syllable structure may differ in surface and non-surface forms arises because surface stop-h sequences are representationally ambiguous (2). On one hand, stop-h sequences could be realizations of a new aspirated stop category (2a) (O’Neill 2009; Gylfadotir 2015). They look quite similar to aspirated stops in languages like English and German, at least on the surface. Under this conceptualization, the syllable preceding the stop has no coda and is light at all levels of representation. On the other hand, stop-h sequences may still be realizations of underlying /sC/ sequences (as they have been diachronically), derived by metathesis (2b). Under this analysis, segmental order and syllable structure may differ in non-surface and surface forms. At some non-surface representational or derivational level, /s/ syllabifies as a coda, creating a heavy syllable. On the surface, however, /s/ (realized as [h]) is no longer in that syllable due to metathesis. The resulting stop-h sequence is the onset to the following syllable, leaving the preceding syllable light (see Section 4.6.2 for further discussion of resyllabification).

(2) Hypothetical representations and syllable structure

- a. Hypothetical /C^h/ representation: syllable structure the same in non-surface and surface forms
 /'tʃip^ha/ ['tʃi.p^ha] (LL)
- b. Hypothetical /sC/ representation that undergoes metathesis: syllable structure differs in non-surface and surface forms
 /'tʃispa/ ['tʃis.pa] (HL) → ['tʃih.pa] (HL) → ['tʃi.pha] (LL)

Most existing research treats stop-h sequences as deriving from /sC/ sequences that undergo metathesis, as in (2b) (e.g., Torreira 2006; Parrell 2012; Torreira 2012; Ruch 2013; Cronenberg et al. 2020). This treatment is also supported by results from a behavioral experiment (Gilbert 2023) in which Sevillian listeners treated stop-h sequences as realizations of underlying /sC/ clusters: they interpreted [h] in word-initial stop-h sequences as /s/ on the preceding word. The abstract representation (/sC/) does not seem to have changed, despite changes in pronunciation towards metathesis ([Ch]).

The current paper tests the weight of syllables preceding stop-h sequences by investigating how they interact with stress. Spanish stress shows weight-sensitive restrictions, one of which is that antepenultimate stress does not occur in words with heavy penults (Harris 1983; Roca 1991). Given that stop-h sequences may arise from /sC/ sequences and involve a change in syllable structure, listeners could treat words with antepenultimate stress where /s/ has metathesized out of the penult differently from those with simple CV penults ('CV.CV.CV vs. 'CV.CV.ChV.). Although they have the same weight profile on the surface (LLL), they may differ at non-surface levels of representation.¹

1.1 The current paper

The current paper has two goals. The first goal is experimental. I present results from a forced-choice stress acceptability task designed to test the weight of syllables that have undergone metathesis (syllables where /s/ has metathesized out). In the task, listeners compare nonce words with antepenultimate stress that have penults of different shapes. The results suggest that listeners strongly prefer words with CV penults over words where /s/ has metathesized out of the penult, and over words with unambiguously heavy penults. A plausible interpretation is that (a) there is a mismatch between the surface form [Ch] and underlying form /sC/; (b) stress is evaluated on a non-surface form that has a heavy penult. I also discuss other possible interpretations of why these syllables may be treated as heavy (Section 4.6)

The second goal is theoretical. Starting from the assumption that stop-h sequences derive from /sC/ clusters, I explore what kinds of theoretical frameworks can account for the interaction between stress and metathesis in Sevillian. The results from the experiment do not distinguish between several possible analyses, so I sketch analyses within two categories. *Process interaction* analyses treat the Sevillian pattern as resulting from the interaction of two phonological processes, stress assignment and metathesis. The interaction is opaque because stress is constrained by structure not visible on the surface. More specifically, it falls into the category of *countershifting opacity* (Rasin 2022). If future research supports these analyses, the experimental results are interesting because they document productive opacity in an experimental setting (contra e.g., Sanders 2003). Process interaction analyses

¹I describe the experiment presented in this paper as testing how metathesis affects syllable structure. This framing already implies somewhat of an analysis: that stop-h sequences are derived from /sC/ sequences and that metathesis results in a change of syllable structure. The basic observation is that there could be a mismatch between the structure over which stress operates and the surface form, which might be visible in listeners' willingness to accept stress in certain locations. This can only be considered a 'mismatch' if stop-h sequences are derived from /sC/ clusters. If stop-h sequences are realizations of underlying aspirated stops, there is no metathesis or process interaction to speak of. The representational status of these sequences is thus directly related to the potential interaction.

need to be either serial or rule-based in order to account for the pattern. Parallel, constraint-based analyses cannot account for the interaction. *Representational* analyses treat stress and metathesis as processes that differ in kind, and are relegated to different representational levels of the phonological grammar such that they do not interact. Representational analyses can take many forms as long as the two processes are separated. I explore a system in which stress operates on layers of representation that include different kinds of timing and gestural information.

This paper is organized as follows. Section 2 provides more information on /sC/ variants and metathesis in Sevillian Spanish, Spanish stress, metathesis and syllable structure, and experimental approaches to weight sensitivity in Spanish. Section 3 describes the stimuli for the main experiment and presents results from small production and perception studies to verify their acoustic properties. Section 4 presents the main experiment and several possible interpretations of the results. Section 5 sketches various theoretical analyses that could account for the interaction. Section 6 concludes.

2 Background

2.1 Spanish coda /s/ and Sevillian metathesis

The production of coda /s/ is variable throughout the Spanish-speaking world. From a categorical perspective, common realizations include the unreduced variant [s], debuccalization to [h], and deletion [Ø] (e.g., Cedergren 1973; Terrell 1979; Lipski 1986; Lipski 1994). These variants are often conceived of as forming part of a reduction continuum, from less to more reduced (3) (e.g., Ferguson et al. 1990; Mason 1994).

$$(3) \quad /VsC/ : \quad (\text{retention}) \quad [Vs.C] \quad [Vh.C] \quad [V\emptyset.C] \quad (\text{deletion})$$

—————→

Coda /s/ reduction has operated diachronically in the development from Latin to Spanish and other Romance languages, and is often considered to reflect a general pressure against coda consonants (Alonso 1945; Malmberg 1965; Mason 1994). Coda reduction is often approached as progressive feature loss: reduction occurs by loss of oral features ([s] → [h]) followed by loss of laryngeal features ([h] → [Ø]) (Goldsmith 1981; Hualde 1989a; McCarthy 2008b). However, recent studies show that /s/ reduces more gradiently than can be captured by discrete allophonic categories (e.g., File-Muriel and Brown 2011; Erker 2012; Henriksen and Harper 2016).

Coda /s/ variants are in synchronic variation in all varieties of Spanish, and varieties differ in which realizations are most common. Common factors that condition this variation include speech rate, speech style, morphological and phonological context, and speaker demographics (see Erker 2012: 11–20 for a thorough overview). Despite high synchronic variation, coda /s/ realizations seem to be stable in most communities (Labov 1994: 583–5). It is possible that coda /s/ is undergoing change, as has historically occurred, but at a rate not visible in synchronic variation patterns (Mason 1994: 72–6).

Sevillian Spanish has the variants just described plus several others, including metathesis. Ruch (2008) reports rates of variants for /st/ clusters in production data from 53 Sevillian speakers in several speech styles (Table 1). Based on impressionistic classification, she finds variants where /s/ remains in its original location ([st], [ht]), variants with metathesis ([th], [ts]), variants where /s/ is phonetically split around /t/ ([hth], [sth]), and variants with /s/ deletion accompanied by variable lengthening of /t/ ([t, t:]). Note the two possible results of metathesis for /st/ sequences: [th] (*post-aspiration*; *stop-h* in my terminology) and [ts] (*post-affrication* in Ruch’s 2008 terminology).

Table 1: Rates of /st/ variants in Sevillian Spanish (Ruch 2008). Most frequent realization per speech style is bolded.

Variant		a. Conversation	b. Reading	c. Word List
Retention	[st]	3.3	17.4	71.3
Debuccalization	[ht]	3.5	3.2	1.5
Surrounding	[hth], [sth]	11	9.5	5.2
Stop-h	[th]	49.1	43.7	13.0
Post-affrication	[ts]	22.0	13.7	6.6
Deletion and/or gemination	[t, t:]	11	12.4	2.4
		100%	100%	100%

In conversational speech (Table 1a), /st/ is realized as metathesized ([th] or [ts]) in over 70% of forms. Other forms, including the pan-Hispanic standard ([s] retention) and non-standard reductions common in other Spanish varieties (debuccalization, deletion), are comparatively infrequent. In read speech, metathesized realizations constitute over 55% of forms, but the proportion of standard [s] realizations increases (Table 1b). Only in the most careful speech, word list reading, are metathesized forms less frequent than non-metathesized ones (Table 1c). Even in that style, the categories with metathesis are the largest after [s] retention. I am unaware of an existing study that presents rates of variants for /sp/ and /sk/ sequences, although phonetic studies show that these sequences also undergo metathesis (e.g., Ruch and Peters 2016; Henriksen, Galvano, and Fischer 2023).

These variants are undergoing change in Andalusian Spanish varieties. From both categorical and gradient perspectives, metathesis (stop-h variants) is advanced in Sevillian Spanish, and most advanced among younger speakers (Ruch 2008; Ruch and Peters 2016). Patterns of stratification by speech style, speaker age, and level of education suggest that the stop-h variant is well-established in Seville and may be a regional standard. In contrast, post-affrication is a newer change. It has received scholarly attention only since the early 2000s (Moya Corral 2007; Ruch 2008; Vida-Castro 2016; Del Saz 2019a; Moya Corral and Tejada Giráldez 2020; Vida-Castro 2022). Between 1995 and 2015, rates of post-affrication increased dramatically in Western Andalusia (Vida-Castro 2022) and may carry covert prestige (Ruch 2008; Vida-Castro 2016; Vida-Castro 2022).

The diachronic progression of these processes appears to have gone as follows: coda /s/ weakening/debuccalization (+ lengthening of the following consonant) → metathesis. For sequences where the second consonant is /t/, there is a further step: → post-affrication (Ruch 2008: 69; Vida-Castro 2016; Del Saz 2019b; Moya Corral and Tejada Giráldez 2020).² It is crucial that debuccalization appears to have preceded metathesis, because debuccalization may be a necessary precursor for articulatory and perceptual reasons (Section 5.1.4). Although metathesis is not new (Moya Corral and Tejada Giráldez 2020 find evidence of sporadic metathesis in a linguistic atlas from the 1960s), coda /s/ weakening occurs in all parts of the Spanish-speaking world and is older, dating from at least the 16th century (Romero 1995a: 192–3). It is not clear whether metathesis is part of the cline of coda reduction and whether it is motivated by the same factors. One proposal is that it is, and that it is a good solution to the pressure against coda consonants: it removes /s/ from coda position while maintaining some of its features (Ruch 2008; Vida-Castro 2016; Moya Corral and Tejada Giráldez 2020; Vida-Castro 2022). However, this proposal is not obviously well supported, especially given that coda reduction and metathesis are often accompanied by lengthening of the stop closure, which can be interpreted as maintenance of the coda timing slot.³

My experimental stimuli use stop-h forms for consistency across /sp, st, sk/ sequences (only /st/ has a post-affricated variant). Furthermore, while both both stop-h and post-affricated variants are common for /st/, [ts] sequences appear to carry more social meaning (and possibly stigma). Stop-h sequences are recognizable as Sevillian (Ruch 2018), but are standard enough to avoid calling excessive attention in an experimental setting.

2.2 Spanish stress

In non-verbs, Spanish stress is uncontroversially contrastive and controversially weight sensitive. Primary stress falls in a right-aligned three-syllable window and can be penultimate ([ka'ðera] ‘hip’), antepenultimate ([saβaðo] ‘Saturday’), or final ([koliβri] ‘hummingbird’). Penultimate stress is by far the most common; antepenultimate stress is infrequent. Bárkányi (2002) presents a study of stress patterns in non-verbs based on the *Diccionario de la Real Academia Española*. Based on my own calculations from her numbers, 9% of nouns have antepenultimate stress, 75% have penultimate stress, and 15% have final stress (roughly similar to my calculations based on numbers reported by Morales-Front 2014: 244).

Stress is usually taken to be lexically marked in non-verbs, which can contrast in the location of stress (e.g. /'sabana/ ‘bed sheet’ vs. /sa'bana/ ‘savannah’) (Harris 1983; Harris 1991). However, several restrictions also suggest weight sensitivity. For example, words with final stress largely end in consonants (Harris 1983; Harris 1991; Roca 1991), and words ending in a consonant restrict the stress window to the final two syllables (Harris 1992: 7). The crucial weight-related restriction for the current experiment is in (4):

- (4) * $\acute{L}H\sigma$: A heavy penult or final syllable prevents antepenultimate stress (Harris

²Sequences where /t/ is the second consonant include /st/, /pt/, and /kt/, and post-affrication is limited to these sequences. /sp/ and /sk/ undergo metathesis but not post-affrication.

³Lengthening of the following closure does not *necessarily* mean that metathesis could not be driven by pressures against codas. For example, the pressure against codas could be seen as a dispreference for Place features that are not linked to an onset (as in Itô 1989). Coda [s] would violate it, but a long consonant that arises by autosegmental spreading from the preceding onset would not.

1983: 85, 88–90; Roca 1991; Harris 1992; Baković 2009; Baković 2016).

The * $\acute{L}H\sigma$ restriction captures the fact that a heavy penult or final syllable narrows the stress window to the final two syllables. There are exceptions, but these tend to be place names or loanwords (e.g. *Mánchester*, *Ámsterdam*; examples from Roca 1990), which arguably do not result from the language’s productive grammar.⁴

The * $\acute{L}H\sigma$ restriction is evident in corpus studies of the Spanish lexicon. Bárkányi (2002) finds several relevant generalizations about non-verbs, shown in Table 2. First, when the final two syllables are light, the most frequent stress location is penultimate (Table 2, row a). Second, when either the penult or ultima is heavy but the other is light, the heavy one is usually stressed (Table 2, rows b, c).

Table 2: Stress patterns in Spanish by syllable type (adapted from Bárkányi 2002: 383).

Structure of final two syllables				Antepen.	Penultimate	Final	Total
a.	...CV.CV	...taka	LL	12.96%	85.68%	1.35%	2261
b.	...CV.CVC	...takan	LH	3.53%	15.26%	81.21%	5184
c.	...CVC.CV	...tanka	HL	.02%	99.38%	.59%	4550
d.	...CVC.CVC	...tankan	HH	.56%	22.4%	77.04%	710

Although antepenultimate stress is infrequent overall, its frequency depends on the weight of the final three syllables. Table 2 (Antepen. column) shows that words ending in two light syllables, LL, have the highest percentage of antepenultimate stress (12.96%). Words with a heavy penult, ultima, or both, very rarely have antepenultimate stress (never constituting more than 4% of words with the given shape). Most importantly, words with heavy penults rarely have antepenultimate stress: .02% of words ending in HL, and .56% of words ending in HH (Table 2, rows c, d, shaded in gray). Fuchs (2018: 9) finds no such words in the corpus of written texts in EsPal, a corpus containing over 300 million word tokens (Duchon et al. 2013).⁵ In short, a heavy penult almost always prevents antepenultimate stress.

While some theoretical proposals for Spanish argue against weight-sensitivity (e.g., Roca 1990; Roca 1991 et seq.; Baković 2016; Piñeros 2016), there is evidence that speakers apply weight-sensitive patterns in experimental tasks in experimental tasks (Section 2.4).

⁴In addition to having rare stress patterns, these words violate other aspects of Spanish phonotactics (e.g., Spanish disallows word-final /m/). The generalization in (4) is also violated by some existing Spanish words (e.g. *régimen* ‘regime’) (Harris 1983; Roca 1990; Baković 2016). These words are difficult for many accounts of Spanish stress, but are exceptions to a very strong tendency.

⁵The numbers reported by Fuchs (2018) appear to exclude loanwords, which I think is due to the way loanwords are transcribed in EsPal. In EsPal, loanwords seem to lack orthographic stress—for example, *Wáshington* is spelled as *Washington*. The automatic data processing described in Duchon et al. (2013) codes stress according to the general rules of Spanish; because stress is not marked orthographically on loanwords, words with heavy penults are coded as having penultimate stress. As far as I can tell, Fuchs’s (2018) search in this database for antepenultimate stress and heavy penults would not have captured these non-native words because they were not marked as having antepenultimate stress.

2.3 Metathesis changes syllable structure

Under the assumptions that (a) Spanish stress is weight-sensitive, and (b) Sevillian stop-h sequences are realizations of abstract /sC/ sequences, stress and metathesis could interact as illustrated in Table 3. When metathesis has not occurred, /s/ is syllabified as a coda, which creates a heavy syllable. Metathesis moves /s/ (debuccalized to [h]) to after the onset to the next syllable, and the sequence syllabifies as an onset. The preceding syllable is light.

Table 3: Metathesis changes surface syllable structure

	No metathesis	Metathesis	
/pasta/	[ˈpas.ta] <u>HL</u>	[ˈpa.tha] <u>LL</u>	‘pasta’
/españa/	[es.ˈpa.ɲa] <u>HLL</u>	[e.ˈpha.ɲa] <u>LLL</u>	‘Spain’
/espeθjalidad/	[es.pe.θja.li.ˈðað] <u>HLLLH</u>	[e.phe.θja.li.ˈðað] <u>LLLLH</u>	‘specialty’

If metathesis changes syllable structure, it potentially creates a mismatch between surface-visible structure and a non-surface structure upon which stress operates. Stress could be conditioned by the phonetic structure visible on the surface, or by structure that is present only at a non-surface level. In contrast, the prediction for words with antepenultimate stress and heavy penults (e.g., CVN, CVS, CVH) are clearer. Given that antepenultimate stress is blocked in words with heavy penults, listeners should disprefer nonce words with this structure. This should hold regardless of the level of representation their decision is based on.

Table 4 illustrates possible decisions listeners could make for words with /CVS/ penults. Listeners should disprefer antepenultimate stress words where /s/ is realized phonetically as coda [s] or [h], because these penults are heavy at both underlying and surface levels. For the metathesized variant, listeners could go either way. They may accept antepenultimate stress words where /s/ has metathesized out of the penult. This would suggest that they treat the penult as light, evaluating stress on the surface form where the penult has no coda. Or, they might disprefer words with antepenultimate stress and these kinds of penults, just like CVS and CVH penults. This would suggest that they treat the penult as heavy, evaluating stress on a structure not visible on the surface, where /s/ is still in coda position.

An alternative explanation for why penults might be treated as heavy is that metathesis does not really change syllable structure. Section 4.6.2 argues against this possibility.

2.4 Perception of stress in Spanish

Recent research argues that weight sensitivity is an active restriction in Spanish, and that it affects participants’ production and acceptance of stress patterns. Participants apply weight sensitivity to nonce words and are sensitive to stress-weight configurations that violate

Table 4: Possibilities for stress judgments in words with /CVS/ penults

Assumed UR	Phonetic re- alization	Evaluation based on UR	Evaluation based on phonetic realization
/CVC <u>V</u> SCV/	CV.CVS.CV	Disprefer antepen. stress + these penults	Disprefer antepen.
	CV.CVH.CV		stress + these penults
	CV.CV.CHV		Accept antepen. stress + this penult

restrictions (Shelton 2007; Shelton, Gerfen, and Gutiérrez Palma 2009; Shelton, Gerfen, and Gutiérrez Palma 2012; Shelton 2013; Shelton and Grant 2018; Fuchs 2018).

For example, Shelton (2007) finds that Spanish speakers are sensitive to unattested combinations of stress location and heavy syllables. He asks Spanish-speaking participants to read nonce words aloud. The nonce words have antepenultimate or penultimate stress, and different kinds of penults. He tests words that (1) are phonotactically illegal, and (2) are phonotactically legal in the synchronic system, but absent for historical reasons. Participants made more mistakes with phonotactically illegal words (e.g., *dóvalda*: antepenultimate stress, heavy penult) than with phonotactically legal words (e.g., *dóvasa*: antepenultimate stress, light penult; *doválda*: penultimate stress, heavy penult). For words that are phonotactically legal but absent for historical reasons, Shelton (2007) used nonce words with palatal onsets to the final syllable (e.g., *dovaña*). These words should allow antepenultimate stress because all syllables are light.⁶ In the experiment, participants produced words with final palatal onsets with error rates between those for phonotactically legal and illegal nonce words. Shelton suggests that speakers are ‘differentially sensitive to what is...theoretically prohibited by the synchronic grammar as well as to structures that are absent for historical reasons’ (Shelton 2007: 101). Weight restrictions in the synchronic grammar play a role in participants’ stress judgments. See also Fuchs (2018) for a study on how Spanish speakers treat words that should permit antepenultimate stress synchronically, but do not for diachronic reasons.

Garcia (2019) targets the distinction between analogy and syllable weight in Brazilian Portuguese, a closely related language with stress patterns similar to Spanish. In a prior lexicon study, Garcia (2017: 68) found that antepenultimate stress is more likely in words with *light* antepenults than in those with heavy antepenults. In a subsequent study, Garcia (2019: 616) found that LLL words are more likely to bear antepenultimate stress (24% of LLL words) than HLL words (21% of HLL words). This finding is unnatural from the perspective of weight sensitivity, since heavy syllables typically attract stress.

Garcia’s (2019) speakers do not generalize this unnatural pattern to nonce words in an experimental setting. His participants compared words with identical segmental material

⁶The historical reasons for their absence are as follows. Palatal nasals arose in Spanish from Latin words that underwent reduction in various ways. In some cases, diphthongization reduced syllable count and led to a palatal nasal: *his.pa.ni.a* > *es.pa.nja* > *es.pa.ña*, ‘Spain’ (Shelton 2007: 80). In Latin, *his-* fell outside the three-syllable window in which stress could fall. Diphthongization reduced syllable count so that the corresponding syllable in modern Spanish, *es-*, falls within the window, but stress did not shift.

that had penultimate vs. antepenultimate stress (*prísbade* vs. *prísbáde*). The words had weight profiles LLL, HLL, LHL, and LLH. Participants chose the penultimate stress version more than the antepenultimate stress version only for LHL words. For LLL, HLL and LLH words, listeners chose the antepenultimate stress version more than the penultimate stress version. The crucial result is that antepenultimate stress is chosen at a higher rate in HLL words than in LLL words: stress is ‘attracted’ to the antepenultimate syllable in HLL words (presumably because it is heavy) at a higher rate than in LLL words. This behavior goes against the lexical statistics above, where LLL words are more likely to have antepenultimate stress than HLL words. It also goes against the pattern observed for the subset of trisyllabic words in the lexicon, where most of the words that have antepenultimate stress are LLL (71%), and many fewer are HLL (29%) (Garcia 2019: 629). If participants’ choices were based on reference to existing forms, they should have chosen the antepenultimate stress version at higher rates in LLL words than in HLL words. Instead, participants applied a more natural pattern of weight sensitivity, with the heavy syllable attracting stress, despite the conflicting pattern in the lexicon.

Finally, Fuchs (2018) found that Spanish speakers apply weight sensitivity restrictions to nonce words in a stress judgment task. The nonce words had light penults ([da.ti.pe.βo]) and various types of heavy penults (e.g., [da.ti.pem.bo]), and each word was presented in written form with both penultimate and antepenultimate stress. Participants gave higher ratings to words with penultimate stress than to those with antepenultimate stress in all of the syllable structure conditions (e.g., [da.ti.‘pe.βo] > [da.‘ti.pe.βo]). Within words with penultimate stress, words with heavy and light penults received similar ratings. However, within words with antepenultimate stress, words with heavy penults received significantly lower ratings than words with light penults ([da.‘ti.pe.bo] > [da.‘ti.pem.bo]). This suggests that the restriction against antepenultimate stress when the penult is heavy is active in Spanish speakers’ synchronic grammars, and is applied productively to nonce words. Heavy penults make antepenultimate stress worse.⁷

My experiment, described in the next section, builds on Fuchs’s (2018) findings and (partially) methodology, applying them to Sevillian Spanish words with syllables whose codas have metathesized out.

3 Experiment set-up

The main research question is as follows: Do Sevillian listeners treat syllables where /s/ has metathesized out as heavy or light? The experiment uses the interaction of metathesis with stress to test this. Metathesis creates a possible mismatch between syllable weight at surface and non-surface levels. Stress could operate on either level, and listeners’ responses to words with metathesis make clear which.

In the main experiment, listeners heard pairs of nonce words with antepenultimate stress that differed only in the type of penult. Section 3.1 describes the stimulus words for the main experiment. Section 3.2 describes the modeling procedures used for these preliminary

⁷Not all experimental studies conclude that weight sensitivity is active in speakers’ grammars. Several of the earlier experimental studies interpret their results as supporting analogy-based stress assignment (Aske 1990; Bárányi 2002; Alvord 2003; Eddington 2004; Face and Alvord 2005; Face 2006). Others argue for roles of both analogy and weight-sensitivity (e.g., Eddington 2004; Waltermire 2004).

experiments and the main experiment. Sections 3.3 and 3.4 present acoustic and perception studies of the stimuli in order to ensure that they were produced, and are perceived, with stress in the intended location. Most of the stimuli are phonotactically illegal in Spanish (antepenultimate stress + heavy penult), and could have been difficult for a native speaker to produce. The results presented in this section show that the acoustic cues to stress are present on antepenultimate vowels in the stimulus words, and that Spanish speakers perceive stress in the intended (antepenultimate) location.

3.1 Stimuli for the main experiment

The nonce words for the main experiment have four syllables and antepenultimate stress. The onsets to the final syllable were /p, t, k/ (which can host metathesized [h]); the vowels in the antepenultimate and penultimate syllable were /a, i, u/. In a given word, the same vowel was used in both the antepenultimate and penultimate syllables (/ginakapo/) to avoid the possibility that listeners' choices were based on the ability of a particular vowel to bear stress.⁸

The nonce words were designed in sets. Table 5 illustrates the three /p/-final onset sets as an example (each column is a set). Members of each set differ only in the type of penult: CV (no coda), CVN (coda sonorant), CVS (coda [s]), CVH (coda [h]), CV.CH (metathesized [h]). There were 45 test nonce words (9 base CV words x 5 penult types). The CV forms of the other sets are listed at the bottom of Table 5.

Table 5: Three of the word sets (/a, i, u/ + /p/-final onset) used as stimuli for the stress judgment task

	/....aCap.../	/...iCip.../	/...uCup.../	CV base words for other sets
CV	gi'nak <u>a</u> po	tʃu'nif <u>i</u> po	na'luf <u>u</u> po	lu'mafato
CVN	gi'nak <u>a</u> mpo	tʃu'nif <u>i</u> mpo	na'luf <u>u</u> mpo	ki'palako
CVS	gi'nak <u>a</u> s <u>p</u> o	tʃu'nif <u>i</u> s <u>p</u> o	na'luf <u>u</u> s <u>p</u> o	la'pikito
CVH	gi'nak <u>a</u> hpo	tʃu'nif <u>i</u> hpo	na'luf <u>u</u> hpo	tʃa'kiliko
CV.CH	gi'nak <u>a</u> pho	tʃu'nif <u>i</u> fpho	na'luf <u>u</u> fpho	ta'mukuto tʃa'lupuko

36 filler words were also included (4 filler words in each of the 9 sets). The filler conditions involved changes that should not affect stress judgments. Two fillers in each set differed minimally from the CV word by (1) changing the onset of the final syllable from a stop to a nasal ([gi'nakapo] → [gi'nakamo]), and (2) changing the voicing, continuancy, nasality, or rhoticity/laterality of the first consonant ([gi'nakapo] → [ki'nakapo]). Another two fillers in each set differed minimally from the CV.CH word by (1) changing the original voiceless stop

⁸Sonority-driven stress has been claimed for some languages, but has also been questioned. See Shih and de Lacy (2019) for overview and arguments against it. Regardless, keeping the vowels the same removes this factor as a potential confound.

final onset to a different voiceless stop ([gi'nakapho] → [gi'nakatho]), and (2) changing the first consonant ([gi'nakapho] → [ki'nakapho]).

The nonce words were recorded by a linguistically-trained, male native speaker of Sevillian Spanish in his late 20s. The recording was done in a quiet room with a Zoom H4N Pro recorder. He was instructed to produce all words with antepenultimate stress and different variants of coda /s/ ([s], [h], [Ch]). Although words with this combination of stress and syllable structure do not exist in Spanish, he produced them easily, as verified in the following acoustic analysis and perception study.

3.1.1 Neighborhood density

Because antepenultimate stress marked in Spanish, listeners might be more willing to accept it on nonce words that are more similar to existing words with antepenultimate stress. I controlled for neighborhood density for the CV base words using Levenshtein distance, which is the edit distance to turn one string into another. Calculations were based on orthography.⁹

First, I created a subset of four-syllable words with antepenultimate stress from SUBTLEX-ESP, a 41 million word Spanish corpus based on film subtitles (Cuetos et al. 2012). Then, I calculated the Levenshtein distance between each CV nonce word candidate and each word in the corpus subset (*stringdist* package for R, van der Loo 2014). Although the Levenshtein distance is a rough measure of neighborhood density, it has been found to explain much of listeners' judgments of word similarity (Vitevitch and Luce 2016: 78).

For the experimental items, I chose CV nonce words that had no lexical neighbors at a distance of three or less, and no more than five neighbors at a distance of four. The goal was to use nonce words that are similar—but not too similar—to real words. Dautriche (2017:fn. 13) suggests that word confusability effects tail off after an edit distance of one to two, so words at a distance of three should be similarly susceptible to effects—or lack thereof—of lexical neighbors. The rest of the words in each set were built off these chosen CV penult words.

3.2 Statistical modeling

Data analysis for the preliminary production and perception experiments, as well as for the main perception experiment, were run in R (v.4.1.3; RCoreTeam 2022). Models were run using *lme4* (v.1.1.27.1; Bates et al. 2015) and *lmerTest* (v.3.1.3; Kuznetsova, Brockhoff, and Christensen 2017). Post-hoc tests were done with *emmeans*, with Tukey adjustments for multiple comparisons (v1.6.2-1; Lenth 2020). Further details about the models are included in the relevant sections.

3.3 Acoustic analysis of stimuli: Stress

Although antepenultimate stress on words with heavy penults is rare in Spanish, an acoustic analysis suggests that my stimulus words were successfully produced with antepenultimate

⁹There are potential confounds to using orthography for calculating neighborhood density, but the orthography-to-phoneme correlation in Spanish is almost one-to-one and was deemed sufficient for these purposes.

stress. Spanish primary stress is realized acoustically with a combination of f_0 , duration, and intensity (Listeri et al. 2003; Ortega-Llebaria and Prieto 2010). Stressed vowels are longer and have slightly higher intensity, but intensity is a weak and inconsistent correlate (Ortega-Llebaria and Prieto 2010). In Castilian Spanish (North-Central Spain), stressed syllables have a rising pitch contour (Ortega-Llebaria and Prieto 2010; Vogel, Anthanasopoulou, and Pincus 2016) that peaks on the following syllable (Ortega-Llebaria and Prieto 2010). I have not found studies of Sevillian intonation, but rising contours on stressed syllables are reported for nearby varieties (Henriksen and García-Amaya 2012).

The stimulus words show acoustic evidence of stress on the antepenultimate vowel, as intended. This acoustic analysis compares antepenultimate vowels ([gi'nakapo]; intended to be stressed) to penultimate vowels ([gi'nakapo]; intended to be unstressed). Figure 1 shows that antepenultimate vowels are longer than penultimate vowels. Figure 2 shows an example of an F0 contour on antepenultimate and penultimate vowels. The antepenultimate vowel, 'u1', has rising F0 that peaks on the following (penultimate, unstressed) vowel. The penultimate vowel, 'u2', has falling F0. In other stimulus words, the contours for the penultimate vowel are flatter, but most fall slightly because of the early peak from the preceding stressed vowel. What is crucial is that the contour differs substantially from the rising contour on stressed vowels. Antepenultimate and penultimate vowels do not differ in intensity.

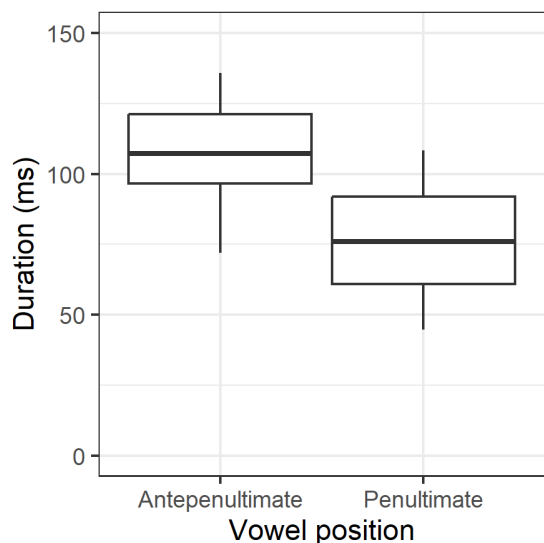


Figure 1: Duration of vowels in stimuli nonce words by position (line = mean, box = 25%-75% IQR; whiskers = 5% and 95%)

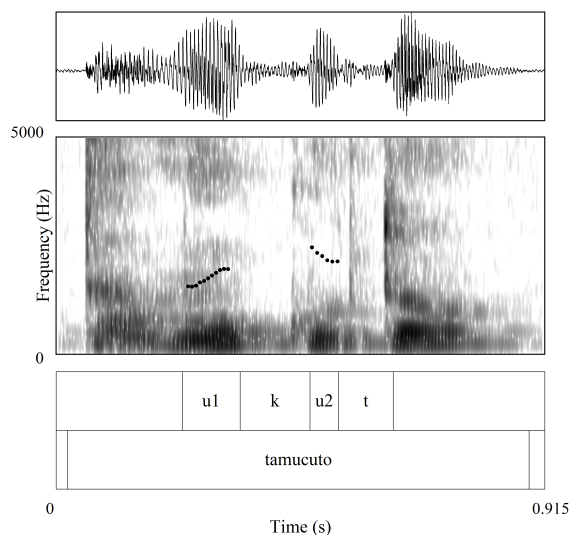


Figure 2: F0 track for antepenultimate and penultimate vowel in [ta'mukuto] (pitch range: 75-250Hz)

Linear regression models support these observations. The Duration and Intensity models contain fixed effects of Word Condition (CV, CVN, CVS, CVH, CV.CH), Vowel Position (antepenultimate, penultimate), and their interaction. Penultimate vowels are shorter than

antepenultimate vowels ($\beta = -.031$, $p < .01$). Neither Word Condition, nor the interaction between Vowel Position and Word Condition, are significant. The difference in duration holds across conditions. This is important because it suggests that penultimate vowels are not shorter simply because many of them come from syllables with a coda consonant, while antepenultimate vowels all come from open syllables. Penultimate vowels are shorter than antepenultimate vowels even when penultimate vowels come from open syllables too (in words with CV penults).¹⁰ The Intensity model had no significant effects.

For F0, measurements were taken at four points during each vowel (20ms, 30ms, 40ms, 50ms). For antepenultimate vowels, the maximum F0 on the *following* (penultimate) vowel was also taken, in order to capture the expected peak on the syllable following primary stress. Each vowel (/a, i, u/) was modeled separately in each position (antepenultimate, penultimate). The models had a fixed effect of Measurement Location. Results reported are from models and post-hoc tests with *emmeans*. For antepenultimate vowels (all qualities), the maximum F0 on the following vowel is higher than F0 at all locations within the antepenultimate vowel ($p < .001$ for /a, i, u/). Although the F0 rise within the antepenultimate vowel is not significant, manual inspection shows that many of these vowels show a rise even within that vowel, as visible in Figure 2. For penultimate vowels, measurement location was not a significant predictor of F0, but there is visually a slight fall in many of the words. This may be because the first measurement point was 20ms into the vowel, failing to capture the full extent of the fall.

In sum, the antepenultimate vowels in my stimuli carry the acoustic correlates of stress: they are longer than unstressed vowels of the same quality, and have rising F0 contours with a peak on the following syllable. That stress is not reflected in intensity is not surprising, since intensity effects in Spanish stress are small, inconsistently produced, and inconsistently used to identify stress (Llisteri et al. 2003; Ortega-Llebaria and Prieto 2009; Ortega-Llebaria and Prieto 2010).

Other acoustic properties of the stimuli could plausibly affect the interpretation of results of the main experiment, specifically the duration of the final onset consonant, of the penultimate syllable rhyme, and the segments involved in metathesis. Further discussion can be found in Section 4.6.2 and Appendix A. In brief, acoustic duration does not strongly correlate with participants' responses in the main experiment.

3.4 Preliminary perception study

This preliminary perception study ensures that Spanish-speaking listeners hear stress on the syllable that was intended to be stressed in the stimulus words for the main study.

¹⁰When vowel quality is included in the Duration model as a third fixed effect and in interaction with Condition and Vowel Position, one effect is marginally significant: the difference between stressed and unstressed /a/ is greater than that of /i/. I leave vowel quality out of the model because, for each vowel quality, the antepenultimate version is longer than the the penultimate one of the same quality (/a/ antepen. = 125ms, pen. = 74ms; /i/ antepen. = 102ms, pen. = 79ms; /u/ antepen. = 94ms, pen. = 75ms).

3.4.1 Materials and task

Recall that the stimuli for the main experiment are four syllables long and have antepenultimate stress (see Table 5). Because they all have antepenultimate stress, it was necessary to modify these words to design a task testing where listeners hear stress.

I created two 3-syllable versions of each nonce word—one with antepenultimate stress and one with penultimate stress—by removing the first and final syllables, respectively ([na'lufupo] → ['lufupo], → [na'lufu]).¹¹ Table 6 illustrates the modified nonce words used in this preliminary perception study.

Table 6: Example stimuli for one word set for preliminary stress experiment

Original	Condition	Stress location	Word created	Answer choices
[na'lufupo]	CV	Penult	[na'lufu]	na <u>lufu</u> , <u>na</u> lufu
		Antepen.	['lufupo]	<u>lufu</u> po, luf <u>u</u> po
[na'lufumpo]	CVN	Penult	[na'lufum]	na <u>lufum</u> , <u>na</u> lufum
		Antepen.	['lufumpo]	<u>lufum</u> po, luf <u>u</u> mpo
[na'lufuspo]	CVS	Penult	[na'lufus]	na <u>lufus</u> , <u>na</u> lufus
		Antepen.	['lufuspo]	<u>lufus</u> po, luf <u>u</u> s <u>po</u>
[na'lufuhpo]	CVH	Penult	[na'lufuh]	na <u>lufus</u> , <u>na</u> lufus
		Antepen.	['lufuhpo]	<u>lufus</u> po, luf <u>u</u> s <u>po</u>
[na'lufupho]	CV.CH	Penult	[na'lufu]	na <u>lufus</u> , <u>na</u> lufus
		Antepen.	['lufupho]	<u>lufus</u> po, luf <u>u</u> s <u>po</u>

The experiment was implemented in PCIBex (Zehr and Schwarz 2018). In each trial, listeners heard one word and clicked on one of two given orthographic representations to indicate the word they heard. The two answer choices differed only in the location of stress. Following orthographic conventions of Spanish, antepenultimate stress was marked with an acute accent and penultimate stress was not marked.¹² Stressed syllables were underlined

¹¹This method produced three-syllable words that are not phonotactically legal in Spanish for the word sets with /p/-final onsets—for example, a word like [na'lufumpo] in the CVN condition is shortened to [na'lufum]. [m] is not a legal coda. I do not believe this is an issue, since the answer choices are identical except for stress: *nalufum* has penultimate stress, and *na'lufum* has antepenultimate stress. According to convention, m-final words should have an orthographic accent to mark penultimate stress, e.g., *nalúfum*. These words in my experiment did not. This lack of orthographically indicated stress on m-final words did not affect listeners' accuracy in any systematic way. The fact that the syllables intended to be stressed were marked in the answer choices may also have mitigated the effect.

¹²Stressed antepenultimate <qui> and <gui> syllables were mistakenly written in the answer choices with the accent on <u> instead of <i>. This mistake affected the words derived from <guinakapo> and <quipalaco>, in the versions created to have penultimate stress. For example, when the audio was [gi.'na.ka], the answer choices were written as <guinaca> (penultimate) and <guínaca> (antepenultimate), instead of the correct <guínaca>. The mistake also affected <chaquilico> in the versions created to have antepenultimate stress (audio: ['kiliko], answer choices <quilico> (penultimate) and <qúilico> (antepenultimate), instead of <quílco>). This mistake does not appear to have lowered accuracy systematically on the three

in both answer choices to ensure that participants visually noticed the difference between choices. Words in three conditions had the same answer choices: CVS, CVH, and CV.CH words all had answer choices with orthographic <s>, even though [s] was not acoustically present in CVH and CV.CH words. For example, in the Penultimate Stress condition, [na'lufuh] ($\leftarrow$ [na'lufuhpo]) and [na'lufu] ($\leftarrow$ [na'lufupho]) had answer choices <nalufus, nálufus>, which would be the corresponding orthographic forms. This should not be problematic since the answer choices differed only in stress and /s/ reduction and deletion in this context are common for this variety.

There were 162 words. Listener groups A and B each heard half of the words. Listeners in Group A heard half of the words in a given set with penultimate stress (CV, CVS, CV.CH, Filler1, Filler3) and the other half of the words in the same set with antepenultimate stress (CVN, CVH, Filler2, Filler4). For the a different word set, they heard each word type with the opposite stress pattern. This pattern repeated for all of the word sets. Listeners in Group B heard the inverse of Group A. Trials were randomized. There were also two practice items and six attention checks.

3.4.2 Participants

20 participants from Spain were recruited on Prolific. 10 were assigned to listener group A and 10 to listener group B. Region within Spain was not controlled, since the goal is to check if Spanish speakers generally hear stress in the intended location. The experiment lasted 5-15 minutes and participants were paid. One listener from Group A and three from Group B were excluded for answering more than one attention check wrong, so the remainder of the analysis includes 16 listeners.

3.4.3 Results

Listeners heard stress where it was intended to be with 88%-98% accuracy across penult type conditions (Figure 3). A mixed-effects logistic regression (fixed effect of Word Condition, random intercepts for Participant and Item) finds one significant effect: accuracy was higher for the Attention Check words than the reference level CV words ($\beta = -1.58$, $p < .05$). This is expected because the attention check items were unambiguous and listeners who failed more than one were excluded. The model and post-hoc tests with *emmeans* found no significant differences between other conditions. Listeners had high overall accuracy in identifying the stressed syllable where it was intended to be, and accuracy did not depend on condition.

In short, the acoustic cues to stress are present on the intended antepenultimate syllables in the stimuli, and Spanish-speaking listeners perceive them. I will thus assume that listeners' behavior in the main experiment is based on a correct perception of stress location.

items affected. That the answer choices also had the stressed syllables underlined may have mitigated the effect.

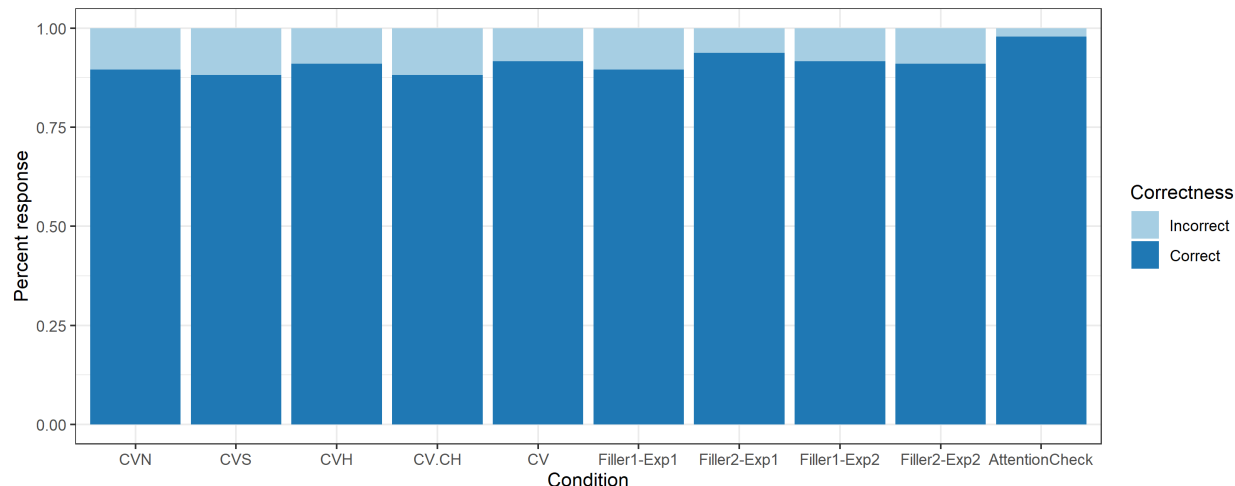


Figure 3: Accuracy identifying stress location in stimulus words, by penult type

4 Main experiment

4.1 Task

Participants completed the task in their homes in 2020 and were asked to wear headphones. In each trial, participants heard two words with antepenultimate stress that differed only in the type of penultimate syllable, paired as in Table 7 (e.g., CV-CVN, [gi'nakapo]-[gi'nakampo]). The CV set compares words with different penult types to the word with a CV penult, testing whether words with CV penults are preferred over those with different kinds of heavy penults and the CV.CH penult. The CV.CH set compares words with different penult types to the word with a CV.CH penult, testing whether words with CV.CH penults are treated differently from words with penults that are unambiguously heavy or light.

Table 7: Condition pairings for CV-base and CV.CH-base comparisons

Comparison type					
CV			CV.CH		
Test	CV	- CVN	CV.CH	- CV	
	CV	- CVS	CV.CH	- CVN	
	CV	- CVH	CV.CH	- CVS	
	CV	- CV.CH	CV.CH	- CVH	
Filler	CV	- Filler1-1	CV.CH	- Filler2-1	
	CV	- Filler1-2	CV.CH	- Filler2-2	

Participants were told that the words were not real words and were asked to choose which word would be a better word of Spanish by clicking buttons on the screen labeled '1' or '2', corresponding to the first and second words they heard.

There were 108 trials, composed of the 9 word sets (Table 5) arranged into the 12 comparisons in Table 7. In each trial, the two words were played with 300ms in between; once the participant made a choice, the experiment advanced automatically. Trial order was randomized for each participant, and the order of presentation of the audio files within each trial was counterbalanced. Group A heard half of the word pairs with the base form (CV or CV.CH) first and the comparison form second (e.g., CV-CVS), and the other half with the comparison form first and the base form second (e.g., CVH-CV). Group B heard the opposite orders. There was one practice trial. Participants also filled out a demographic questionnaire and were paid. The experiment lasted 12-35 minutes.

The task was inspired by Fuchs (2018), but differs in several key ways. First, I present the stimulus words auditorily instead of orthographically, which allows me to test different allophonic realizations of the same word. Given that metathesis is not represented orthographically, this is crucial. Additionally, Fuchs' task was a goodness-rating task (using a 5-point Likert scale), which resulted in significant, but small, effects. The current study forces participants to make a binary choice in an attempt to distill their preferences.

4.2 Hypotheses

In general, listeners are expected to prefer words with light penults over those with heavy penults. Table 8 lays out listeners' expected behavior if they treat CV.CH penults as heavy (Table 8a) or light (Table 8b). For the CV-base comparisons, listeners should prefer CV words over alternative CVN, CVS, and CVH words. This is the expected preference under both sets of hypotheses, since CV.CH words are not involved. The CV-CV.CH comparison distinguishes the two hypotheses. If listeners treat the CV.CH penult as heavy, they should prefer CV > CV.CH. If they treat the CV.CH penult as light, they should not have a preference between CV and CV.CH.

The pairs in the CV.CH comparisons test the hypothesis more directly by comparing CV.CH words with words of other penult types. For the CV.CH-base comparisons, the two hypotheses predict different listener behavior for each comparison type. If listeners treat CV.CH penults as heavy, they should have no preference between CV.CH and CVN, CVS, and CVH words, and they should prefer CV words over CV.CH words. (This last preference is the same as the CV-CV.CH comparison in the CV-base comparisons). If listeners treat CV.CH penults as light, they should have no preference between CV.CH and CV words, and should prefer the CV.CH word over CVN, CVS, and CVH words.

There are multiple possible explanations for why the penults created by metathesis might be treated as heavy or light. In order to separate the experimental results from the interpretation of those results, I leave further discussion of these possibilities for Section 4.6.

4.3 Statistical models

The results were modeled in mixed-effect logistic regressions using *glmer* from *lme4* in R, and *emmeans* was used to determine estimated marginal means and contrasts between other levels of the main predictor (same details as in Section 3.2). Separate models were built for the CV and CV.CH comparisons, predicting the likelihood of choosing the Base form (CV or CV.CH) over the Alternative. The dependent variable was the response of the Base

Table 8: Predictions for listener responses based on whether they treat CV.CH penults as heavy or light

Base	a. CV.CH penult: heavy	b. CV.CH penult: light	Alternative
CV	>	>	CVN
	>	>	CVS
	>	>	CVH
	>	=	CV.CH
CV.CH	<	=	CV
	=	>	CVN
	=	>	CVS
	=	>	CVH

form (coded as 1) vs. the Alternative (coded as 0). Coefficients of logistic regressions are in log odds. Positive coefficients indicate a higher log odds of Base response; lower coefficients indicate lower log odds of the Base response (and thus higher log odds of the Alternative response). A log odds value of 0 corresponds to a probability of .5, while positive log odds correspond to probability $>.5$ and negative log odds correspond to a probability of $<.5$.

Table 9 shows the fixed effect (Comparison Type) and random intercepts (Word Set, Participant) in the models. A random slope of Condition by Participant and Condition by Word Set were not included because models fit with them resulted in singular fit warnings. Comparison Type is dummy coded, and the underlined Comparison Type is the reference level for the factor. The models respond to the following question. There is a certain log odds of a Base response (e.g., CV) in the reference level of Comparison Type (e.g., CV-CVN). Is the log odds of a Base response in another Comparison Type (e.g., CV-CVS) significantly different from the log odds of a Base response in the reference level Comparison Type?

Table 9: Predictors in models (reference level underlined)

	Predictor	CV model	CV.CH model
Fixed effect	Comparison type	<u>CV-CVN</u>	<u>CV.CH-CVN</u>
		CV-CVS	CV.CH-CVS
		CV-CVH	CV.CH-CVH
		CV-CV.CH	CV.CH-CV
Random intercepts	Word set		
	Participant		

Estimated marginal means (with *emmeans*) are calculated for each Comparison Type based on the model results, and statistical tests are applied to test for significant differences between the factor levels. For easier interpretation, I used a back-transformation within

emmeans to calculate the estimates on the response scale instead of the model’s logit scale. Estimates and confidence limits (95% confidence level) are thus given in probabilities, and contrasts are calculated on the log odds ratio scale. The estimated probabilities and confidence limits for each Comparison Type calculated from *emmeans* are overlaid on the plots of my listeners’ data.

4.4 Participants

The participants were 27 Sevillians (20 female, 7 male), with an average age of 37.9 years (range: 18-70). They were born and completed schooling in the city of Seville and smaller towns in the province. Most had lived their entire lives in the region, although some reported short-term (<1 year, UK, León [Spain], Majorca [Spain]) and long-term stints (2+ years, Switzerland/Finland, US, Madrid [Spain], Galicia [Spain]) elsewhere. All participants had completed high school, thirteen had completed a technical or university degree, and twelve had done postgraduate studies (one did not report education). Participants reported knowledge of languages other than Spanish: English (20), French (7), German (2), Italian (2), Portuguese (2), Gallego (1) and ‘Catalan/Valencian’ (1). Three participants reported almost-native proficiency in another language (English, French, Gallego). Some also participated in the experiment reported in Gilbert (2023).

4.5 Results

Figure 4 shows results for the CV comparisons (Figure 4a) and CV.CH comparisons (Figure 4b). The y-axis is the proportion of Base response (CV or CV.CH) over the Alternative. Each dot represents an individual speaker’s response rate for the Comparison Type. The line at .5 is for visual clarity. Dots above the line indicate that a listener chose the Base form over the Alternative more than 50% of the time. Dots below the line indicate that a listener chose the Alternative over the Base more than 50% of the time. The blue dots and error bars are predicted probabilities and 95% confidence limits produced by *emmeans* based on the logistic regressions.

4.5.1 CV comparisons

In the CV comparisons (Figure 4a), the overall rate at which listeners chose CV over the Alternative ranged from 67%-74% across Comparison Types (percentages calculated within-category, over all listener responses). Listeners preferred words with light penults over each Alternative (CV > CVN, CVS, CVH, CV.CH; [gi'nakapo] > [gi'nakampo], [gi'nakaspo], [gi'nakahpo], [gi'nakapho]). The rate of preference for CV is similar for all of the Alternative forms, including those with surface-heavy penults and those with CV.CH penults.

The logistic regression supports these observations (Table 10). Recall that the model predicts the log odds of choosing the Base CV word over the Alternative. Positive coefficients indicate higher log odds of a CV response in the given Comparison Type than in CV-CVN (the reference level Comparison Type), while negative coefficients indicate the opposite. For CV-CVN comparisons, listeners chose the Base CV word with a log odds of .964 (the intercept; $p < .0001$). In terms of probability, this is an estimated probability of .72 of a CV response.

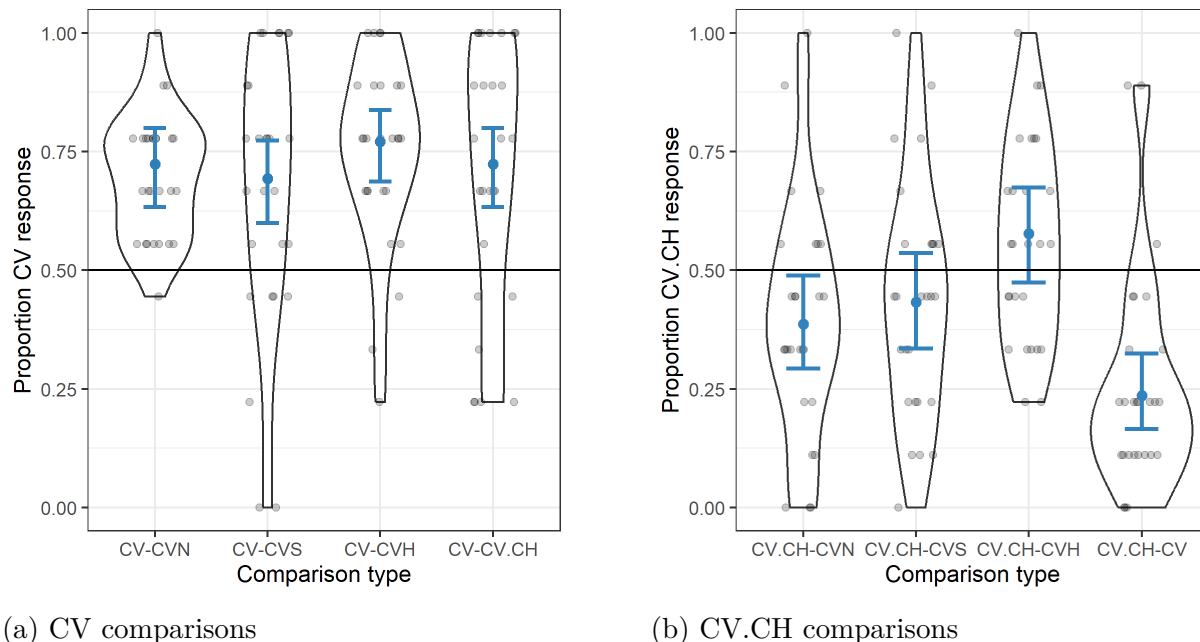


Figure 4: Listener results from the stress judgment task.

The effect of Comparison Type is not significant: the log odds with which listeners prefer the Base CV word over the Alternative are not statistically different between the reference level CV-CVN and any other level (vs. CV-CVS: $\beta = -.149$, $p = .469$); vs. CV-CVH: $\beta = .250$, $p = .240$; vs. CV-CV.CH: $\beta = 0.000$, $p = 1.000$).

The predicted probabilities for each level of Comparison Type (from *emmeans*) are in Table 11. For each Comparison Type, the predicted probability of a CV response is well above chance (ranging from .69-.77). Furthermore, the 95% confidence limits do not span chance (.5), indicating a high likelihood that the true parameter lies above chance. Listeners choose the Alternative over the base form significantly over chance in all comparisons. The *emmeans* contrasts between the predicted probabilities suggest that the probabilities are not significantly different between any of the Comparison Types (results not shown).

Table 10: Model for CV comparisons predicting probability of CV response (vs. Alternative)

	Estimate	Std. Error	z value	Pr(> z)
Reference: CV-CVN				
(Intercept)	0.964	0.213	4.519	0.000***
CV-CVS	-0.149	0.205	-0.725	0.469
CV-CVH	0.250	0.212	1.176	0.240
CV-CV.CH	0.000	0.208	0.000	1.000

That listeners prefer CV over CV.CH words at a similar rate as they prefer CV over CVN, CVH, and CVS words does not suggest a robust difference between conditions, but does not provide positive evidence for their similarity. The next set of comparisons tests CV.CH

Table 11: *emmeans* predictions by Comparison Type for CV model

Comparison Type	prob	SE	Lower CL	Upper CL
CV-CVN	0.724	0.043	0.633	0.799
CV-CVS	0.693	0.045	0.599	0.773
CV-CVH	0.771	0.039	0.687	0.838
CV-CV.CH	0.724	0.043	0.633	0.799

Confidence level used: 0.95

Intervals are back-transformed from the logit scale

words directly against CVN, CVS, and CVH words to address this issue. Do listeners treat all penult types as similarly unacceptable? The statistically significant results in the CV.CH comparisons are compatible with the lack of effect in the CV comparisons.

4.5.2 CV.CH comparisons

In the CV.CH comparisons (Figure 4b), the overall rate at which listeners chose CV.CH over the Alternative was less than 50% in CV.CH-CVN comparisons (40%) and CV.CH-CVS comparisons (44%). They dispreferred CV.CH words in comparison to CVN and CVS words ([gi'nakapho] < [gi'nakampo]; [gi'nakapho] < [gi'nakaspō]). Listeners chose the CV.CH word even less frequently in CV.CH-CV comparisons (26%), which had a CV word as the Alternative ([gi'nakapho] < [gi'nakapo]). In contrast, the CV.CH-CVH comparisons were the only ones in which listeners chose the CV.CH word at rate over 50% (56%; ([gi'nakapho] > [gi'nakahpo])).

The logistic regression (Table 12) supports the visual interpretation of the results. In CV.CH-CVN comparisons (the reference level of Comparison type), the log odds of a CV.CH response is -0.463 ($p < .05$). This indicates that listeners are less likely to choose CV.CH than CVN (log odds -.0463 is a probability of .386 of choosing CV.CH). In relation to this reference level, the log odds of choosing CV.CH in CV.CH-CVS comparisons is not significantly different ($\beta = .193$, $p = .325$). However, in comparison to the reference level, the log odds of a CV.CH response is significantly lower in CV.CH-CV comparisons ($\beta = -.710$, $p < .001$) and significantly higher in CV.CH-CVH comparisons ($\beta = .774$, $p < .0001$).

Table 12: Model for CV.CH comparisons predicting probability of CV.CH response (vs. Alternative)

	Estimate	Std. Error	z value	Pr(> z)
Reference: CV.CH-CVN				
(Intercept)	-0.463	0.214	-2.167	0.030*
CV.CH-CVS	0.193	0.196	0.984	0.325
CV.CH-CVH	0.774	0.196	3.939	0.000***
CV.CH-CV	-0.710	0.210	-3.385	0.001***

The *emmeans* estimates of the probability of a CV.CH response for each Comparison Type are shown in Table 13. In most Comparison Types, the predicted probability of a CV.CH response is $<.5$ (lower than chance). CV.CH-CVH comparisons are the only ones for which the predicted probability of a CV.CH response is $>.5$. Note that the 95% confidence limits include chance (.5) for CV.CH-CVS and CV.CH-CVH comparisons, indicating that we cannot exclude the possibility that the true estimate could be at chance. The upper confidence limit for CV.CH-CVN comparisons is also close to .5 (.489). The *emmeans* contrasts between the predicted probabilities (Table 14) suggest that the predicted probabilities are significantly different between all levels of Comparison Type, except CV.CH-CVN vs. CV.CH-CVS.

Table 13: *emmeans* predictions by Comparison Type for CV.CH model

Comparison Type	prob	SE	Lower CL	Upper CL
CV.CH-CVN	0.386	0.051	0.293	0.489
CV.CH-CVS	0.433	0.052	0.335	0.536
CV.CH-CVH	0.577	0.052	0.474	0.674
CV.CH-CV	0.236	0.041	0.166	0.325

Confidence level used: 0.95

Intervals are back-transformed from the logit scale

Table 14: *emmeans* contrasts between levels of Comparison Type for CV.CH model

contrast	odds.ratio	SE	p.value
(CV.CH-CVN) / (CV.CH-CVS)	0.82	0.16	0.7588
(CV.CH-CVN) / (CV.CH-CVH)	0.46	0.09	0.0005***
(CV.CH-CVN) / (CV.CH-CV)	2.03	0.43	0.0040**
(CV.CH-CVS) / (CV.CH-CVH)	0.56	0.11	0.0151*
(CV.CH-CVS) / (CV.CH-CV)	2.47	0.52	0.0001***
(CV.CH-CVH) / (CV.CH-CV)	4.41	0.93	<.0001***

P value adjustment: tukey method for comparing a family of 4 estimates

Tests are performed on the log odds ratio scale

Although there is a general dispreference for CV.CH words when compared to words with CVN and CVS penults (a point further discussed in Section 4.5.3), there are two crucial points. First, the predicted confidence limits for the CV.CH-CVS comparison includes chance (.335-.536); for CV.CH-CVN comparisons, the confidence limits approach chance (.293-.489). Second, the predicted probability of a CV.CH response in CV.CH-CV comparisons is significantly lower than in all other comparisons, and the confidence limits are well below chance (.166-.325). In other words, despite an overall dispreference for CV.CH in

comparison to some words with heavy penults, there is a significantly *stronger* dispreference for these CV.CH words when the alternative has a CV penult.

4.5.3 Results discussion

Table 15 summarizes the results. Listeners prefer antepenultimate stress words that have CV penults over similar words that have closed penults (CVN, CVS, CVH) or penults where /s/ has metathesized out (CV.CH). Regardless of what consonant or surface manifestation of /s/ closes the penult, Sevillian listeners are more likely to choose the CV word and do so at similar rates over the alternatives. When choosing between antepenultimate stress words that have CV.CH penults vs. those that have CVN, CVS, or CVH penults, listeners have preferences, but they are weak. They choose CV.CH words less frequently than CVN and CVS words, and more frequently than CVH words, although the confidence limits around these predicted probabilities include, or are near, chance.

The crucial point is the following: despite the weak dispreference for CV.CH words when the alternative words are CVN or CVS, this dispreference for CV.CH words is much stronger when the alternative is a CV word. The relative strength of this dispreference suggests that CV.CH words are treated more similarly to words with surface-heavy penults than to words with light CV penults. The results are broadly as predicted if CV.CH penults are treated as heavy (as in Table 8).

Table 15: Summary of listeners' preferences in the stress judgment task

CV Comparisons			CV.CH Comparisons		
CV	>	CVN, CVS, CVH, CV.CH	CV.CH	~/<	CVN, CVS
			CV.CH	~/>	CVH
			CV.CH	<	CV

The differences between rates of preference for the CV.CH word when compared to words with CVN, CVS and CVH penults are somewhat unexpected. The CVN, CVS, and CVH words all have heavy penults and should be treated similarly. Furthermore, regardless of whether CV.CH forms have heavy or light penults, neither predicts that they should be dispreferred to forms with surface-heavy penults like CVN and CVS—they should either be treated similarly to these alternatives (if treated as heavy) or preferred over them (if treated as light). The CV.CH < CVN, CVS dispreference does not necessarily undermine the results. The dispreference for CV.CH could actually further support the idea that CV.CH forms are treated as having heavy penults, since they are even more dispreferred than words with unambiguously heavy penults.

One result does not fit clearly with the predictions: listeners prefer CV.CH over CVH words, and this is the only comparison where they choose CV.CH over 50% of the time. If CV.CH derives from /sC/, CV.CH and CVH forms share an underlying representation and should be treated similarly. I suspect the preference for CV.CH over CVH words arises because this comparison is qualitatively different. The judgment between CV vs. CVN words is a decision between two different words. The judgment between CV.CH vs. CVH words

is a decision between allophonic realizations of the *same* word. Between two non-standard pronunciations, they choose the one that is more frequent in their dialect (CV.CH) (see Section 2.1). Thus, this preference could be due to their familiarity with surface phonetic forms, rather than to differing phonological structure of the forms. Another possibility is that the acoustics of the CVH stimulus words were particularly bad: a combination of surface-present coda [h] and a long closure duration of the final onset consonant resulted in an acoustically over-long penult (measurements given in Appendix A).

If participants prefer CV.CH over CVH because CV.CH is the more common allophonic variant of /sC/ sequences, we might expect also expect listeners to prefer CV.CH over CVS, which is also infrequent in conversational speech (Section 2.1). They do not: CV.CH forms are chosen slightly *less* than 50% of the time in CV.CH-CVS comparisons. I speculate that this is because the full sibilant variant is the national and international standard. While listeners prefer the reduced form of their own dialect over reduced forms that are rare in their dialect (CV.CH > CVH), they do not always prefer the reduced form of their dialect over the standard (CV.CH $\approx$ / < CVS). The fact that CV.CH was preferred over CVH but not over CVS could also be due to the fact that words with CVS penults did not have the acoustic ‘over-heaviness’ of the CVH penults (Appendix A).

These observations lead to a possibly different interpretation of the results: that listeners’ choices between the words were based on sociolinguistic factors, not syllable weight.¹³ Listeners’ preference for CV words in CV-CVN and CV-CVS comparisons would be based on the heaviness of CVN and CVS penults. In CV-CVH and CV-CV.CH comparisons, however, listeners’ preference for CV words could be because CVH and CV.CH are nonstandard variants. The additional surface weight of the CVH word would compound the dispreference for CVH words. This explanation could also apply to the slight CV.CH < CVN, CVS dispreference: CV.CH words contain a nonstandard variant and the others do not. When evaluating this possible explanation, it is important to keep in mind that although CV.CH realizations of /sC/ sequences are non-standard at the supraregional level, they appear to be somewhat of a regional standard. In data from nearby Málaga (Vida-Castro 2016), CV.CH variants are common in most read speech styles, are increasingly frequent as speaker age increases, and are most common in the speech of university-educated people. (These patterns contrast with post-affricated variants, which are most frequent among young speakers and speakers with lower levels of education, and are avoided in read speech.) Whether these explanations hold would come down to how listeners interpreted the task: if their decisions were based on the phonotactics of a new lexical item, or on the presence of (non)standard allophonic variants. This would require further empirical testing.

4.6 Results interpretation: Syllable weight

This constellation of results suggests that CV.CH penults are treated as heavy when it comes to stress assignment, similar to CVS, CVN and CVH penults. There are several possible ways to interpret these results. Because there is no evidence that the syllables created by metathesis are treated as light, I focus only on different reasons they may be treated as heavy: opaque process interaction or representational separation (Section 4.6.1 and Section

¹³Thanks to Eric Baković for suggesting this possibility.

5), syllabification (Section 4.6.2), and acoustic duration (Section 4.6.3).

4.6.1 Process interactions and representational separation

That CV.CH penults are treated as heavy is in line with conceptualizing stress and metathesis as processes that apply serially, in an opaque interaction where either stress is assigned before metathesis or the two processes apply at the same time to the same representation. In either case, stress only ‘sees’ the heavy penult, not the result of metathesis. Another way of understanding the stress-metathesis ‘interaction’ is as representational separation: stress and metathesis operate on different levels of the phonological grammar, on different types of representations, so the result of one cannot affect the result of the other. The goal of Section 5 is to flesh out some of these analytical possibilities in more detail.

4.6.2 Syllabification

Another way of interpreting the result that Sevillians seem to treat CV.CH penults as heavy would be to say that these penults maintain heaviness due to syllabification. Stop-h sequences could syllabify across the syllable boundary, with the stop in the coda of the preceding syllable (VC.H, heterosyllabic), as opposed resyllabifying as an onset (V.CH, tautosyllabic). Another possibility is that the stop is representationally linked to both syllables. I discuss these issues here so readers can take them into consideration in interpreting the results, but syllabic affiliation can only be established with future experimental work.

Resyllabification: If stop-h forms are syllabified as VC.H, metathesis would result in the same syllable structure post-metathesis as pre-metathesis, but with the segments swapped (*[gi.'na.kah.po], LLHL vs. *[gi.'na.kap.ho], LLHL). Both have a heavy penult. This VC.H syllabification is possible, but unlikely for several reasons.

First, the VC.H syllabification is not compatible with restrictions on coda obstruents in Spanish. Coda obstruents are restricted both word-medially and word-finally, and those that are allowed are often reduced or deleted (Campos-Astorkiza 2012; Colina 2012). This pressure against coda obstruents is synchronically and diachronically evident throughout Romance languages (recall (3)) (Malmberg 1965; Vennemann 1988; Mason 1994 and references therein). Syllabifying the sequence as VC.H maintains a consonant in coda position, and replaces /s/ with a voiceless stop. Depending on the theoretical approach taken—for example, a sonority-based one—a voiceless stop can be even worse than [s] or [h].¹⁴

Second, if /p, t, k/ were syllabified into coda position, they should reduce like other coda obstruents in Western Andalusian Spanish. In these varieties, coda /p, t, k/ behave like coda /s/: they undergo debuccalization and metathesis (and affrication), and their weakening is accompanied by variable lengthening of the following consonant (/apto/ → [at(:)ho] ~ at(:)(s)o) (Del Saz 2019a; Henriksen, Galvano, and Fischer 2023). In Sevillian, /p, t, k/ in metathesized stop-h sequences do not reduce (5), suggesting that they are onsets, not codas (Gerfen 2001).

¹⁴If metathesis were to be driven by the pressure against codas, the VC.H syllabification is also incompatible with this motivation.

- (5) Expected reduction of /p, t, k/ under VC.H syllabification of stop-h sequences
 /turista/ → [tu.rih.ta] → [tu.rit.ha] → *[turih.ha] ‘tourist’

If VC.H is unlikely, the other option is V.CH. One possible objection to the V.CH syllabification is that sequences of stops plus [h] are unattested in Spanish. I follow the discussion in Gilbert (2023) about this issue. Neither the VC.H and V.CH syllabifications create preferred sequences in Spanish. The V.CH syllabification is bad because it creates unattested onset sequences. Additionally, stop-h onsets do not have a sufficient sonority rise (Sonority Sequencing Principle; Clements 1990; Blevins 1995). The VC.H syllabification is bad because it (a) violates the restriction against coda obstruents, as already discussed, and (b) violates the preference for sonority to fall across a syllable boundary (Syllable Contact Law; Murray and Vennemann 1983; Clements 1990; Gouskova 2004). Given that coda obstruents are so dispreferred in these varieties of Spanish, it is not immediately clear whether the VC.H or V.CH syllabification is worse. One theoretical possibility that might ameliorate the bad onset issue with V.CH is Steriade’s (1994) proposal for Mazatec. In brief, she suggests that complex onsets are less marked if they are structurally similar to single segments. In Sevillian, CH onsets may be less marked than they appear because the sequence consists of a closure and release, two phases which can be present in a single segment (in aspirated stops). In short, how to account for CH as a viable onset is an unresolved issue, but it is not obvious that the V.CH syllabification is worse than VC.H.

Multiple-linking: CV.CH syllables also could be interpreted as heavy if the stop is representationally linked to both syllables. Coda /s/ reduction is often accompanied by variable lengthening of the following consonant in varieties of Andalusian Spanish (e.g., Alvar 1955; Romero 1995b; Gerfen 2002; Campos-Astorkiza 2003; Martinez-Gil 2012). Variable lengthening is also reported in varieties with metathesis (e.g., [pat(:)ha]) (Ruch 2008; O’Neill 2010; Ruch and Harrington 2014; Henriksen, Galvano, and Fischer 2023). Long stop closures accompanying /s/ reduction have been treated as compensatory lengthening in multiple varieties of Spanish (Hualde 1989b; Campos-Astorkiza 2003; Martinez-Gil 2012).

A moraic analysis of compensatory lengthening would go as follows (e.g., Hayes 1989). /s/ reduces by delinking (in part or in full) from its mora, leaving the mora slot empty. The following consonant then spreads into that timing slot and is doubly linked, occupying both coda and onset position. Double linking is often argued to manifest as longer acoustic duration (Broselow, Chen, and Huffman 1997; Cohn 2003; Khat tab and Al-Tamimi 2014). Under this representation, the long stop would be syllabified across the syllable boundary (Maddieson 1985), the syllable preceding the stop-h sequence would be heavy, and the assumption that metathesis changes syllable structure would be untenable. While this representation of consonant lengthening could be viable, there are several complications. Lengthening is variable, is inversely correlated with the duration of metathesized [h], and long stop closures appear to be shortening as the metathesis change advances (Parrell 2012; Torreira 2012; Ruch and Harrington 2014). Furthermore, lengthening can occur with /s/ debuccalization, not only deletion, so the moraic slot is not entirely empty. It is not clear how a mora sharing account of lengthening would account for these subtleties (see Gerfen 2002), or the interaction between lengthening and metathesis.

In the stimuli used for the experiment reported in this paper, the duration of final onset consonants in CV.CH and CV words is similar ([gi.na.ka.pho] = [gi.na.ka.po]), so acoustic

lengthening would not provide reason for listeners to treat CV.CH penults as heavier than CV penults. (Appendix A, Figure 1).

4.6.3 Acoustic duration

Finally, it is possible that CV.CH penults were treated as heavy because these syllables were acoustically long in the stimulus words, in a way that made them more acoustically similar to CVN, CVS and CVH words than to CV words. As mentioned in Section 4.6.2, stop closure lengthening ([ginaka(m/s/h)|p|ho]) could contribute to the duration of the penultimate syllable. Other durations that could plausibly affect the interpretation of syllable weight include the Intervocalic Interval from the beginning of the penultimate vowel to the onset of the vowel in the final syllable ([ginak|a(m/s/h)ph|o]) (Steriade 2008; Steriade 2012). Longer duration could be interpreted as ‘heaviness’.

Measurements of the stimulus words and their correlations with listener response rates do not suggest that listeners’ responses in my experiment were straightforwardly based on heaviness coming from acoustic duration. Full details are in Appendix A.

5 Analyzing stress and metathesis in Sevillian

The analyses in this section start from the following assumptions: Sevillian stop-h sequences derive from /sC/ sequences by metathesis, and stop-h sequences resyllabify as the onset of the following syllable.¹⁵

The main observation from the experiment is that syllables where /s/ has metathesized out are treated as heavy. There is a mismatch between an abstract, non-surface form relevant for stress placement, and the surface form. Listeners apply this mismatch productively to nonce words. To capture my listeners’ behavior, the analysis needs to treat CV.CH penults like CVN, CVS, and CVH penults, blocking antepenultimate stress on all even though they differ in syllable structure on the surface. These all receive penultimate stress instead. The analysis also needs to treat CV.CH penults differently from CV penults, blocking antepenultimate stress in the former but allowing it in the latter. Table 16 summarizes these desiderata.

Table 16: Patterns to account for in the analyses

Penult type		Antepenultimate stress permitted?	
CV	gi.na.ka.po	✓	gi.'na.ka.po
CVN	gi.na.kam.po	✗	gi.'na.kam.po (✓ gi.na.'kam.po)
CVS	gi.na.kas.po	✗	gi.'na.kas.po (✓ gi.na.'kas.po)
CVH	gi.na.kah.po	✗	gi.'na.kah.po (✓ gi.na.'kah.po)
CV.CH	gi.na.ka.pho	✗	gi.'na.ka.pho (✓ gi.na.'ka.pho)

¹⁵Analyses based on other syllabifications could be worked out. As syllabification is an empirical unknown at this point, I make these assumptions in order to limit the scope of analyses covered here.

These observations can be analyzed in several types of frameworks. My experimental results do not distinguish between them, so I sketch two main types of analysis.

One type of account is based on *process interactions*: stress and metathesis are phonological processes that interact with each other (Section 5.1). Under this conceptualization, the interaction is opaque because the factors that condition stress assignment are not visible on the surface. Metathesis has obscured them. Furthermore, applying one process before the other changes how the other applies. Stress and metathesis can also be applied at the same time, to the same representation, as long as the output of metathesis is not visible to stress. If these accounts were to be found to be accurate in future research, the experimental results would be significant because they document an opaque pattern that speakers apply productively to novel words. These accounts can vary along two axes: whether they are serial or parallel, and whether they use rules or constraints. The only combination that cannot produce the stress-metathesis interaction is a parallel, constraint-based one.

Another type of account is *representational*: stress and metathesis are fundamentally different types of operations that occur at different levels of representation (Section 5.2). There is no ‘interaction’ and thus no opacity; the apparent interaction falls out from the separation between representational levels in the phonological grammar.

I close the section by evaluating properties of the analyses presented (Section 5.3), and suggest that a promising approach is one that separates stress and metathesis representationally, and where metathesis operates on gestures. A model of this sort could capture variability in metathesis that is both extensive and constrained, and where variation in the degree of acoustic metathesis does not affect stress.

5.1 Process-interaction analyses

5.1.1 SPE

In a rule-based, SPE-style analysis, rules are ordered and apply sequentially, giving rise to intermediate derivational levels (Chomsky and Halle 1968; Kenstowicz and Kisseberth 1979). Phonological processes interact by creating, removing, or changing the context for other rules to apply (or failing to). This account is serial, and rules operate on segments.

This analysis treats stress, /s/ debuccalization, and metathesis as processes that occur due to phonological rules. The Stress rule (6) is a greatly simplified cover rule in order to focus on process order.¹⁶ The simplifying assumptions for Spanish are as follows (see Section 2.2): penultimate stress is default and antepenultimate stress is lexically marked, Spanish stress is weight sensitive, and the location of stress on the surface is determined by mapping an abstract prominence mark to a surface form (Goldsmith 1976; McCarthy and Pruitt 2013). The Stress rule assigns stress in the lexically-marked location unless the penult is heavy, in which case the heavy penult is stressed. If there is no lexical stress, stress is realized in the default location (penult). The Debuccalization rule (7) reduces coda /s/ to [h] before another consonant in the same word or a word boundary. The Metathesis rule (8) changes the order of [h] and [p, t, k] when they are adjacent.

¹⁶Readers interested in Spanish stress assignment can consult a variety of approaches in e.g., Harris (1983), Harris (1991), Roca (1991), and Baković (2016) (non-exhaustive).

- (6) Stress: Realize lexically-marked stress, except if penult is heavy, in which case it must be stressed. (If no lexically-marked stress, stress falls on the penult.)
- (7) Debuccalization: $[s] \rightarrow [h] / \text{ ______ } \{-\text{syllabic or } \#\}$ (Hammond 1989)
- (8) Metathesis: $[h] + [p, t, k] \rightarrow [p, t, k] + [h]$ (modeled off Chomsky and Halle 1968: 361)

Table 17 illustrates the interaction between rules using a nonce word from my perception experiment.¹⁷ Derivations (a)-(c) illustrate the rule interaction for CV, CVN, and CV.CH penults, where stress precedes metathesis. Derivation (d) illustrates how the opposite ordering gives the incorrect result.¹⁸

Table 17: Stress precedes metathesis (a)-(c); metathesis preceding stress gives incorrect result (d)

Stress > metathesis (correct)			
	a. CV	b. CVN	c. CV.CH
	/gi'nakapo/	/gi'nakampo/	/gi'nakaspo/
Stress	gi.'na.ka.po	gi.na.'kam.po	gi.na.'kas.po
Debuccalization	—	—	gi.na.'kah.po
Metathesis	—	—	gi.na.'ka.pho
Output	gi.'na.ka.po	gi.na.'kam.po	gi.na.'ka.pho
Metathesis > stress (incorrect)			
	d. CV.CH		
	/gi'nakaspo/		
Debuccalization	gi.na.kah.po		
Metathesis	gi.na.ka.pho		
Stress	gi.'na.ka.pho		
Output	*gi.'na.ka.pho		

When stress precedes metathesis (Table 17a-c), words with CV penults are treated differently than those with CVN and CV.CH penults. When stress applies, CV penults are light and antepenultimate stress can surface. In contrast, CVN and CV.CH penults are heavy when stress is assigned and antepenultimate stress is thus blocked. By the time metathesis applies to CV.CH words and makes the penults light, stress has already been determined.

¹⁷By definition, nonce words do not have lexical representations or lexical stress. However, my listeners only heard these words with antepenultimate stress and likely would have interpreted it as 'lexically marked' because it could not arise from default stress patterns. The fact that they accept it to differing degrees depending on the shape of the penult is the relevant point.

¹⁸Whether *debuccalization* occurs before or after stress is not crucial, but I group it with metathesis because they are related segmental processes.

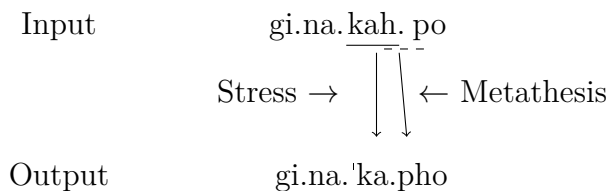
In short, CV words are treated differently than CVN and CV.CH words because CV penults have different syllable structure than CVN and CV.CH penults when stress applies. In contrast, ordering metathesis before stress (Table 17d) fails to block antepenultimate stress in CV.CH words, because the penult is light when stress applies. This order wrongly predicts that CV.CH and CV penults should pattern the same.

Under this analysis, the interaction between stress and metathesis is opaque because metathesis obscures the structure that constrains stress. The results of my perception experiment would be explained as follows: participants evaluated stress on a non-surface form of the word, where the penult was heavy because metathesis had not yet applied.

5.1.2 Simultaneous application

The stress-metathesis interaction in Sevillian can also be derived by applying the Stress rule (6) and Metathesis rule (8) to the same input representation simultaneously.¹⁹ In (9), the Stress rule sees the heavy penult (solid underline), and stress is thus constrained to the penult. The Metathesis rule sees the [hp] sequence in the same input (dashed underline) and switches the order of [h] and [C]. The output has a light penult due to metathesis, but stress was constrained to the penult by the heavy syllable that is no longer visible.

(9) Simultaneous application of stress and metathesis in Sevillian



The fact that simultaneous rule application gives the correct result for Sevillian suggests that serialism—and intermediate derivational representations—are not necessary to analyze this interaction. Indeed, both older and recent work suggests that some opaque interactions can be analyzed with simultaneous rule application instead of serial application (Joshi and Kiparsky 1979; Kenstowicz and Kisseberth 1979:Ch.8; Joshi and Kiparsky 2005; Kiparsky 2015; Pruitt 2023; Baković and Blumenfeld 2024). Section 5.1.5 further discusses the role of serialism in deriving opaque interactions.

5.1.3 Harmonic Serialism

The Sevillian stress-metathesis interaction can be accounted for in Harmonic Serialism (Prince and Smolensky 1993; McCarthy 2010), a serial implementation of Optimality Theory with derivational steps. At each step, the optimal candidate is chosen and passed to another round of constraint evaluation against a new candidate set. The constraint ranking remains the same throughout. Violations are calculated based on the input to the current step, not in relation to the original input (McCarthy 2010). The derivation converges when no further

¹⁹Thanks to Eric Baković for bringing this potential analysis to my attention. This analysis starts from an intermediate representation, but the intermediate representation only sets the conditions for stress and metathesis to apply; it is not necessary to analyze the opaque interaction itself.

changes improve the output and the constraint ranking returns an output identical to the input. HS derivations are gradual: possible candidates at a derivational step are limited to those that differ from the input by a single change, which I define as a faithfulness violation (McCarthy 2007; McCarthy 2008c). The order in which processes apply is determined by constraint ranking. High-ranked markedness constraints force a process to occur early, because any candidate violating them is eliminated in early rounds of evaluation. After this process occurs, satisfying the relevant constraints, other processes that are triggered by lower-ranked markedness constraints can apply.

I assume that changes in syllabification do not ‘count’ as a change for the gradualness requirement because syllabification is not contrastive (not marked in the lexicon) (Clements 1986; Hayes 1989; Blevins 1995; McCarthy 2003). Changes in syllabification thus cannot violate faithfulness constraints.

Furthermore, stress, debuccalization, and metathesis are all single changes that must occur in separate derivational steps. While debuccalization and metathesis clearly violate faithfulness constraints on feature deletion and segmental order, stress deserves further discussion. Stress assignment counts as a change because it violates a faithfulness constraint. I loosely follow McCarthy and Pruitt (2013)’s treatment of stress and metrical structure in HS (see also Goldsmith 1976 for the inspiration behind this idea). Contrastive stress is marked with a diacritic in the underlying representation. This diacritic is not visible on the surface until the phonological grammar maps it to an output. Its surface location is determined by the ranking of violable constraints. Mapping the diacritic to output stress violates a faithfulness constraint such as DEP-STRESS, which penalizes a stress present in the output that has no input correspondent (see Alderete 2001a; Alderete 2001b; Elfner 2016 for similar ideas). In my analysis, this constraint evaluates the input-output correspondence of *stress*, not of the diacritic. I omit DEP-STRESS from the tableau: stress is always realized in the output, despite violations of this constraint. Furthermore, I assume that stress does not shift once it is assigned (Pruitt 2010).

For stress in Sevillian, I use the constraints in (10-12). ID-STRESS (10) prefers stress to be realized on the segment that bears the diacritic in the input.²⁰ HAVESTRESS (11) prefers words to have stress. WEIGHT-TO-STRESS (12) prefers heavy syllables to be stressed.

- (10) ID-STRESS: Assign * for a segment that bears a stress diacritic in the input, that is not stressed in the output.
- (11) HAVESTRESS: Assign * for a word without stress (similar to e.g., Elfner’s [2016] PWDHD).
- (12) WEIGHT-TO-STRESS (WSP): Assign * for an unstressed heavy syllable (Prince 1990).

I assume that coda consonants are assigned moras through Weight-by-Position (Hayes 1989), which requires codas to be moraic.²¹ The details of this process are tangential to the

²⁰The diacritics used by McCarthy and Pruitt (2013) are *h* (*head*) and *d* (*dependent*), and constraints map these into head and dependent positions of metrical feet. Foot structure is tangential to my analysis; I use the diacritic only as an anchor for input-output mappings.

²¹This can be done with violable constraints; see Rosenthal and van der Hulst (1999).

analysis, so the output is shown in Step 1 of the derivation, above the tableau.

Debuccalization to [h] and metathesis are obtained by the interaction of the constraints in (13)-(15). CODA COND penalizes place features in coda position. I treat debuccalization as the result of delinking the oral place feature [CORONAL] from /s/ (Goldsmith 1981; Hualde 1989a). Delinking the place feature satisfies CODA COND (McCarthy 2008b) because only the placeless laryngeal feature [SPREAD GLOTTIS] (realized acoustically as [h]) remains. As counterpoints to CODA COND, MAX(PLACE) penalizes deletion of a place feature and MAX penalizes deletion of a segmental node. For metathesis, *HC disprefers sequences of a glottal fricative plus /p, t, k/, and LINEARITY penalizes changes in segmental order.

- (13) CODA COND: Assign * for a place feature not linked to an onset (Itô 1989; McCarthy 2008b). (Abbreviated CC)
- (14) MAX(PLACE): Assign * for a place feature in the input not present in the output.
- (15) MAX: Assign * for deletion of a segment (root node) (McCarthy 2008b).
- (16) *HC: Assign * for a [hp, ht, hk] sequence.²²
- (17) LINEARITY: Assign * for two elements where the output order [does not] preserve the order of input elements (paraphrased from McCarthy 2008a: 198).

²²The constraint *HC can be justified in several ways (Gilbert 2023). HC sequences seem to be avoided in multiple ways cross-linguistically: (1) coda /h/ is much rarer than onset /h/; (2) aspirated stops are most commonly realized with postaspiration ([Ch]) as opposed to preaspiration ([hC]). Constraints against [hC] sequences arising from aspiration on stops or metathesis, have been proposed by Casserly (2012), Canfield (2015), and Takahashi (2019), among others. The second point draws a parallel between HC sequences that are representationally preaspirated stops vs. /h/ + /C/ clusters, and between postaspirated stops vs. /C/ + /h/ clusters. Several formal analyses argue that there is no representational difference between complex segments and clusters when it comes to aspirated stops and clusters containing a laryngeal and a consonant (e.g., Thráinsson 1978; Kehrein and Golston 2004). If there *is* a representational distinction between (pre/post)aspirated stops and HC and CH clusters, we could still expect similarity: [h] occurs in the same position on the surface regardless of representation, and should be subject to similar auditory constraints (Helgason 2002: 41).

- (18) HS analysis: stress precedes metathesis, blocking antepenultimate stress

Step 1 | **Prosodification:** /gin*akasp/ → [gi.n*a.kas_μ.po]

#	2	gi.n*a.kas _μ .po	HAVESTR	WSP	ID-STR	CC	MAX	MAXPL	*HC	LIN
	a.	gi.na.kas _μ .po	*!	*	*	*				
	b.	gi'.na.kas _μ .po		*!		*				
	c.	gi.na'.kas _μ .po			*	*				
	d.	gi.na.kah _μ .po	*!	*	*			*	*	
#	3	gi.na'.kas _μ .po								
	e.	gi.na'.kas _μ .po				!*				
	f.	gi.na'.kah _μ .po						*	*	
	g.	gi.na'.kap _μ .so				*!				*
#	4	gi.na'.kah _μ .po								
	h.	gi.na'.kah _μ .po							*!	
	i.	gi.na'.ka.ph _μ o								*
	j.	gi.na'.ka.po					*!			

Output:

[gi.na'.ka.pho]

Derivation: /gin*akasp/ → [gi.n*a.kas_μ.po] →[gi.na'.kas_μ.po] → [gi.na'.kah_μ.po] →[gi.na'.ka.ph_μo] (→ [gi.na'.ka.pho])

Step 1, Prosodification, shows syllabification and mora assignment. Coda /s/ has received a mora because it is syllabified as a coda. The output of this step is the input to Step 2.

In Step 2, possible candidates include those with stress (b, c), without stress (a, d), or a single other change from the input (d; debuccalization). A candidate with both stress assigned and debuccalization is not available, since each process is a change and must be done in separate steps. Stress must be assigned at this step: candidates that apply another process first are eliminated by high-ranked HAVESTRESS. Candidate (c) wins this step. It satisfies HAVESTRESS because the surface form has stress, and satisfies WSP because that stress is realized on the heavy penult. In satisfying these high-ranked constraints, (c) violates both ID-STRESS, because the vowel marked for stress in the input is unstressed in the output, and CODA COND, because the place feature of /s/ is not linked to onset position. The other candidates lose by not having stress (a, d), or by faithfully mapping stress to the light antepenult when the penult is heavy (b) (satisfying ID-STRESS but violating WSP).

Step 3 takes (c) from Step 2 as its input. Possible candidates based on this input include one that is identical (e), one with debuccalization (f), and one with metathesis (g). All the candidates shown have stress on the heavy penult, as determined in the previous step. Candidate (e), which was optimal at Step 2, is no longer optimal compared to the new candidate set. Because stress has been assigned, coda /s/ reduction can occur. Candidate (f) wins by debuccalizing coda [s] to [h], satisfying CODA COND by deleting the place feature

of /s/. Only the glottal feature, realized as [h], is left. The *HC violation incurred by debuccalization does not affect the outcome at this step, because satisfying CODA COND is more important. Candidate (e) does not debuccalize, and loses due to its CODA COND violation. Candidate (g), metathesis without debuccalization, would create a [p.s] sequence and is ruled out by CODA COND. Coda [p] is as bad as coda [s] because both have a place feature. A candidate with both debuccalization and metathesis is not available yet, due to the gradualness requirement.

At Step 4, available candidates metathesize [h] (i) or delete [h] (j). Metathesis is preferred over deletion due to the ranking of MAX » LINEARITY, so (i) wins. Note that one more step would technically be needed before the derivation converges: the mora associated with coda /s/ would need to delete after [p] and [h] syllabify as an onset. Onsets are not usually considered to be moraic (e.g., Hayes 1989). Mora deletion would violate a constraint preventing mora deletion, presumably ranked below LINEARITY. At this point, no further changes would be harmonically improving, and the derivation would converge.

This analysis would require further elaboration to account for the acceptability of [Ch] as an onset sequence, the acoustic trade-offs in duration between /s/ and /C/, the range of permissible variability in metathesis, and the motivations of metathesis. In this analysis, debuccalization is driven by constraints on coda consonants, but metathesis is not.

It is notable that the process order necessary to derive the Sevillian stress-metathesis interaction can be analyzed in HS, unlike other opaque interactions (e.g., counterbleeding). Rasin (2022) points out that HS can analyze some types of opaque interactions: counter-shifting ones (discussed in Section 5.1.5). The Sevillian stress-metathesis interaction is of this type.

5.1.4 Parallel Optimality Theory cannot derive the interaction

The Sevillian stress-metathesis interaction is not derivable in parallel Optimality Theory (Prince and Smolensky 1993). The tableau in (19) uses the same constraints and ranking as the Harmonic Serialism analysis, but gives the wrong result. Candidate (c) is the desired winner, but (f) wins. (f) is an undesired winner because it has antepenultimate stress in a word where /s/ has metathesized out of the penult, a pattern that participants in my stress judgment task dispreferred.

(19) Parallel OT analysis cannot derive Sevillian stress-metathesis interaction

	gi.n*a.kas _μ .po	HAVESTR ₁	WSP	ID-STR	CODA COND ₁	MAX	MAXPL	*HC	LIN
a.	gi.na.'kas _μ .po			*!	*				
b.	gi.na.'kah _μ .po			*!			*	*	
⊕ c.	gi.na.'ka.pho			*!			*		*
d.	gi.'na.kas _μ .po		*!		*				
e.	gi.'na.kah _μ .po		*!		*	*		*	
⊗ f.	gi.'na.ka.pho						*		*

All candidates shown have stress. Candidates (a-c) have penultimate stress and heavy penults. They satisfy WSP but are eliminated because they violate ID-STRESS. Candidates (d-e), with antepenultimate stress and heavy penults, satisfy faithfulness to underlying stress location but are eliminated by WSP because the heavy penults are unstressed.

The issue with a parallel OT analysis is that it simultaneously evaluates candidates that are segmentally identical but have stress in different locations. Compare (c) and (f), which both have debuccalization and metathesis and differ only in stress location. Candidate (c) (penultimate stress) is the *desired* winner, but this candidate cannot win when compared with (f) (antepenultimate stress). Candidate (f) is the *actual* winner, because it satisfies ID-STRESS by being faithful to the location of the stress diacritic in the input. In order to correctly prefer (c), the weight of the penult needs to be more important than maintaining faithfulness to underlying stress. However, syllable weight cannot do this in parallel OT: WSP is inactive for both candidates because it sees only surface forms, and both candidates have light penults on the surface. More generally, candidate (c) is harmonically bounded by (f), so no ranking of these constraints would prefer (c) over (f). One could make (c) win by adding a high-ranked constraint that prefers (c) over (f), but any such constraint would be ad-hoc, at best. Harmonic Serialism avoids this issue (Section 5.1.3) through intermediate derivational steps: two candidates with metathesis that differ only in stress location are never in direct competition.

In short, parallel OT is unable to account for the dispreference of antepenultimate stress with CV.CH penults from the experimental results, because the grammar evaluates only surface forms. When only surface forms are evaluated, CV.CH penults pattern with CV penults and differently from CVN, CVS, and CVH penults. This is the opposite of participants' preferences in the experiment.

There is also a segmental issue that affects both this analysis and the HS analysis: a candidate with metathesis but no debuccalization would win if it were syllabified as an onset [Cs]. Imagine two candidates similar to (c) and (f) but without debuccalization: [gi.na.'ka.pso] (c') and [gi.'na.ka.pso] (f'). (c') and (f') would be preferred over (c) and (f), respectively, because they avoid violating MAX(PLACE). The winner of the whole tableau in (19) would be [gi.'na.ka.pso] (f'). Constraints on onset clusters could be added to deal with these forms, but they would need to allow [Ch] onsets and disallow [Cs] ones (see also Section 4.6.2). This distinction could probably be made using sonority sequencing, but [h] and [s] do not obviously differ in sonority (Parker 2002). Furthermore, using constraints on onset phonotactics misses the substantive connections between debuccalization and metathesis. Both may be responses to the pressure for coda reduction (see Section 2.1 for caveats). Or, debuccalization may have been a prerequisite for metathesis. It seems plausible that metathesis only became a possibility once the metathesizing segment had no place features to interfere. Recall that metathesis appears to have arisen after debuccalization diachronically (Section 2.1). Metathesis can arise from changes in the gestural timing of [hC] sequences that occur in real-time speech production (Parrell 2012), and may also be tied to perceptual factors involving [h] in [hC] sequences (Blevins and Garrett 2004; Ruch and Harrington 2014). These factors both could lead to progressively more extensive metathesis over time, and require /s/ to have debuccalized to [h] first.

The parallel OT analysis misses possible connections between debuccalization and metathesis. The [Cs] candidate is required to be possible alongside all the others, because of richness

of the base. HS captures more of the connection, in that debuccalization to [h] can serially precede metathesis. In the tableau in (18), [.Cs] would become available at Step 3 (alongside the [C.s] syllabification already shown). For HS to capture a connection between debuccalization and metathesis, the constraints would need to compel metathesis in a way that makes it possible only when [h] is the metathesizing segment.

The [.Cs] candidate does not make sense given the diachrony and mechanics of metathesis, but once it is considered, it is difficult to rule out with constraints on syllable structure or sonority sequencing. I leave this as a task for future theoretical research.

5.1.5 Comparison of analyses: Opaque interactions

Sections 5.1.1 through 5.1.3 presented analyses that treat stress and metathesis in Sevillian as two processes that interact opaquely: the heavy penult limiting stress placement is not visible on the surface because metathesis has made that syllable light. This pattern can be analyzed through SPE-style rules (rule-based, serial), simultaneous rule application (rule-based, non-serial), and Harmonic Serialism (constraint-based, serial). The interaction cannot be derived in parallel OT (constraint-based, non-serial).

The viability of an analysis for the stress-metathesis interaction may depend more on whether it uses rules or constraints than on whether it is serial or parallel. This observation is inspired by Pruitt (2023), who points out that a framework’s ability to derive opacity depends more on how phonological processes are motivated (rules vs. constraints) than on serialism.

Rule-based frameworks (serial and non-serial) can derive opacity because processes are motivated by input configurations, apply if their conditions are met, and do not evaluate the outputs they produce. These properties are helpful for several reasons. That rules are motivated by input configurations is essential, because opaque interactions are those in which the structure conditioning the application of a processes is present in the input, but is not visible on the surface. That rules do not evaluate their own outputs is essential, because opaque interactions produce forms that violate language-general phonotactics. Rules can be implemented serially or in parallel, achieving the same goal. Serial application involves strict ordering: one process cannot ‘see’ the output of the other, because the second has not applied yet. Simultaneous application applies rules to the same input at the same time so that each is blind to the output of the other. Opacity may intuitively seem to require intermediate derivational steps or serialism, but simultaneous rule application can analyze some kinds of opacity, such as countershifting, without either (Baković and Blumenfeld 2024). For Sevillian stress-metathesis interactions, the crucial point is that rules apply to the input configuration ([hC]) and do not evaluate their output ([Ch]), and that both serial and non-serial approaches prevent stress from operating on the output of metathesis.

Constraint-based frameworks (serial and parallel) generally struggle to derive opacity because constraints motivate phonological processes by evaluating the outputs of these processes. Phonological processes apply in order to repair dispreferred structures, satisfying markedness constraints. Because opaque forms violate phonotactic generalizations of a language (Pruitt 2023: 513), they are ruled out by the same markedness constraints that apply across the language. And because constraints evaluate only the output, they do not ‘see’ the representations that motivate the surface form. This is the main issue for Parallel OT, which

fails to derive opacity because constraints cannot motivate processes based on structure that is not surface-visible. For Sevillian, a parallel OT analysis fails because stress cannot be restricted by the penult that is light on the surface, but heavy at some other level. Even serial, constraint-based frameworks like HS (which give access to intermediate derivational forms) fare almost as poorly as parallel OT in analyzing opacity (McCarthy 2007), due to the nature of constraint evaluation (Pruitt 2023). Despite working with intermediate representations, the markedness constraints still evaluate only outputs at each step.

Although many opaque patterns cannot be analyzed in HS, the stress-metathesis interaction in Sevillian can be. This is because it falls into a category of opaque interactions referred to as *countershifting* (Rasin 2022).²³ In the more commonly discussed opaque interactions, counterfeeding and counterbleeding, the application of one process determines *whether* the second applies, leading to apparent over- and underapplication of rules. In contrast, countershifting results in misapplication, changing *how* another process applies (Rasin 2022; Baković and Blumenfeld 2024). Shifting and countershifting interactions have long been present in the literature (e.g., Joshi and Kiparsky 1979; Kenstowicz and Kisseberth 1979:Ch.8; Zwicky 1987; Joshi and Kiparsky 2005), but are less often recognized as a fundamentally different type of opacity. Serialism allows constraint-based frameworks to handle countershifting opacity in a way that it does not help with counterfeeding and counterbleeding (Rasin 2022). The crucial difference is that in countershifting, the first process changes the context for the other to apply, but does not remove that context. The second process simply applies differently than it would have, had it applied first.

The tableaux in (20) and (21) provide schematic analyses of why countershifting interactions can be modeled in HS while counterbleeding ones cannot. They show a single derivational step: the one that decides which of the two processes occurs first. The tableau in (20) illustrates countershifting with Sevillian stress and metathesis. I show a simplified analysis starting from debuccalized [h], to isolate process order. At the step shown, candidate (a) applies stress, satisfying HAVESTRESS and violating *HC. Candidate (b) metathesizes, satisfying *HC but violating HAVESTRESS. Candidate (a) wins, so stress applies and metathesis will apply at a later step. Crucially, both candidates (a) and (b)—one where stress applies first and the one where metathesis applies first—violate one of the top two markedness constraints. This is the difference from counterbleeding.

(20) HS succeeds at countershifting opacity

	gi.n*a.kah _μ .po	HAVESTRESS	*HC	LIN
☞ a.	gi.na.'kah _μ .po		*	
b.	gi.na.ka.pho	*!		*

²³Baković and Blumenfeld (2024) and older literature (e.g., Zwicky 1987) refer to countershifting as *transfusion*.

(21) HS fails at counterbleeding opacity

	/ħa:kim-in/	*ki	*iCV	MAX
☹ a.	ħa:k ⁱ i.min		*	
☹ b.	ħa:k.min			*

Compare Sevillian countershifting to the counterbleeding interaction between palatalization and vowel deletion in Bedouin Hijazi Arabic (Tableau in (21); adapted from Rasin 2022, example 8). /k/ palatalizes before /i/ (/k/->[k^j]), and high vowels in non-final open syllables delete. Palatalization must occur before deletion, because it applies even when there is no surface-present [i] to trigger it: /ħa:kim-in/ → [ħa:k^jm-in]. The constraint *ki penalizes [ki] sequences (motivating palatalization), and *iCV penalizes high vowels in open, non-final syllables (motivating [i] deletion). For palatalization to occur before deletion, *ki must outrank *iCV. Candidate (a), with palatalization, satisfies *ki but violates *iCV. This candidate needs to win at this step, because palatalization needs to occur before deletion removes the triggering environment. Candidate (b) has deletion but no palatalization. By deleting [i], candidate (b) *violates neither markedness constraint*. *iCV is satisfied because the vowel is deleted. *ki is *also* satisfied, because the constraint is no longer applicable ([ki] no longer exists). Due to its failure to violate *ki, candidate (b) incorrectly wins. At the next step, palatalization cannot be triggered because [i] is already gone.

Several cases of opacity that do not fit squarely into existing diagnostics can be categorized as countershifting opacity. These include stress-epenthesis interactions (Elfner 2016 and Rasin 2022's reanalysis of her data) and a case of vowel copy epenthesis (data from Kuroda 1967 and Steriade 1986; reanalyzed in Baković and Blumenfeld's 2024 example 8). The Sevillian stress-metathesis interaction is another case of countershifting, and appears to be formally very similar to stress-epenthesis interactions.

5.2 Representational analyses

Under a different kind of analysis, what appears to be an 'interaction' between stress and metathesis in Sevillian falls out from these phenomena occurring at different levels of the phonological grammar and on different kinds of representations. Stress operates on a different level and type of representation than metathesis, and metathesis is not visible at the level where stress operates. These accounts involve segmental and gestural representations, so I first introduce a gestural description of Sevillian metathesis (following Parrell 2012; Torreira 2012; Cronenberg et al. 2020).

The gestural score for an /st/ sequence realized as [st] is in (22). [s] has a constricted tongue tip gesture ([critical], producing frication) and a laryngeal gesture (producing voicelessness). /t/ consists of a tongue tip closure gesture. The laryngeal gesture extends slightly past the release of the closure, leading to a short VOT typical of unaspirated voiceless stops.²⁴ Other variants of /st/ sequences involve modifications of these gestures. Debuccalization (23)

²⁴Voiceless stops /p, t, k/ may not have their own open glottal gestures in languages like Spanish (Beckman, Jessen, and Ringen 2013), but adjacent segments often share laryngeal specifications. /t/ does not require voicing, so there is no issue with glottal opening from /s/ continuing throughout.

occurs when tongue tip gesture of /s/ weakens or deletes (gray dashed box), leaving only the glottal gesture (McCarthy 1988; O’Brien 2012).²⁵ Metathesis (24) occurs by gestural realignment: the stop closure gesture shifts leftwards, so that it starts at the same time as the laryngeal gesture (Ruch and Peters 2016; Cronenberg et al. 2020). The acoustic result is that [h] surfaces mostly after the stop instead of before it. Incomplete forms ([hCh]) have also been reported (Ruch 2008; Ruch and Harrington 2014), and are easily captured in this framework as an intermediate phase in the gestural shift.

(22) Retention: /st/ → [st]

Tongue Tip	<table><tr><td>critical</td><td>closed</td></tr></table>	critical	closed	
critical	closed			
Glottis	<table><tr><td>open</td></tr></table>	open		
open				
Acoustic	<table><tr><td>s</td><td>t</td><td>h</td></tr></table>	s	t	h
s	t	h		

(23) Debuccalization: /st/ → [ht]

Tongue Tip	<table><tr><td>critical</td><td>closed</td></tr></table>	critical	closed	
critical	closed			
Glottis	<table><tr><td>open</td></tr></table>	open		
open				
Acoustic	<table><tr><td>h</td><td>t</td><td>h</td></tr></table>	h	t	h
h	t	h		

(24) Metathesis: [ht] → [th]

Tongue Tip	<table><tr><td>critical</td><td>closed</td></tr></table>	critical	closed	
critical	closed			
Glottis	<table><tr><td>open</td></tr></table>	open		
open				
Acoustic	<table><tr><td>h</td><td>t</td><td>h</td></tr></table>	h	t	h
h	t	h		

5.2.1 Representational separation: Walker and Proctor’s (2019) model

Walker and Proctor (2019) present a model in which processes are relegated to different representational levels and operate on different kinds of representations. Phonological optimization is non-serial; it operates over a single level of representation. This level is distinct from other, still-abstract levels that contain more information about how the first level of representations is implemented articulatorily. Representational models can succeed for Sevillian stress and metathesis because stress and metathesis are calculated over different representations. My analysis sketch follows Walker and Proctor’s (2019), but the idea of separating levels of representation to account for properties of phonological processes is also the focus of work by Hall (2006) and Zsiga (2000).

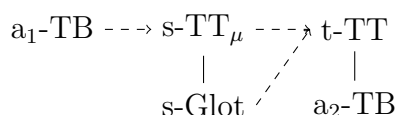
Walker and Proctor’s 2019 framework is based in Articulatory Phonology and has three levels of representation. Level 1 is the **linguistic gestural representation** (the gestural coupling graph), which consists of gestures for the articulators involved, the goal state for the articulators (e.g., closure, narrow, wide; Goldstein, Byrd, and Saltzman 2006), and the temporal coordination relations between the gestures (simultaneous, sequential). Level 2 is the **gestural score**, which specifies the activation period for the gestures and their timing

²⁵This description of debuccalization is similar in spirit to abstract featural description in Section 5.1.3, where debuccalization is conceptualized as delinking the supralaryngeal place feature [CORONAL], leaving only the laryngeal feature.

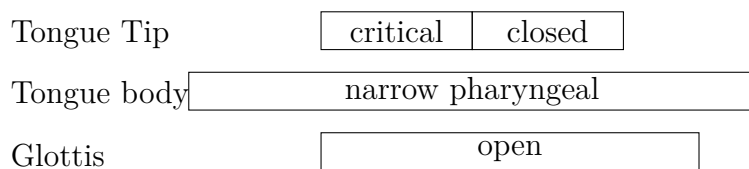
in relation to others, based on the gestural coupling graph from Level 1. Level 3 is the **articulatory trajectories**. These are calculated using the gestural score as input to the Task Dynamic Model, which models temporal and spatial aspects of articulator movement. This third level is still abstract: the actual articulations used to produce the utterance, and the resulting acoustics, are affected by other linguistic and extra-linguistic factors. The model also deals with syllable weight: gestures in specific timing relations are assigned a mora.

For Sevillian, I illustrate the three representational levels with the minimal sequence /asta/ (25-27).

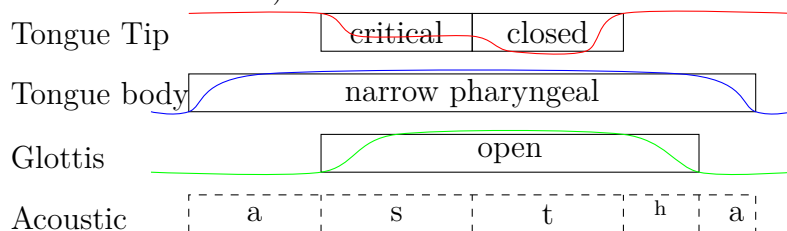
- (25) Level 1: Linguistic gestural representation for /a₁sta₂/



- (26) Level 2: Gestural score for /a₁sta₂/



- (27) Level 3: Articulatory trajectories for /a₁sta₂/ (higher curve = more open; lower curve = more constriction)



The linguistic gestural representation (25) contains gestures and their coupling relations. Vowels /a₁/ and /a₂/ are created by a Tongue Body (TB) gesture (Browman and Goldstein 1990). /s/ and /t/ have the same gestures and goal states as in (22). The coordination relations assumed are standard in Articulatory Phonology. A consonant is timed synchronously with a following vowel in the same syllable ([as.ta]) (in-phase, solid lines). A vowel is timed sequentially with a coda in the same syllable ([as.ta]) (anti-phase, dotted lines) (Browman and Goldstein 1988; Goldstein, Byrd, and Saltzman 2006; Nam, Goldstein, and Saltzman 2009). I also assume that consonant sequences across syllable boundaries are timed sequentially ([as.ta]). For Sevillian, the tongue body gesture of /a₁/ is coordinated sequentially with the tongue tip gesture of /s/. Due to this sequential timing relation, /s/ is moraic.²⁶

²⁶The glottal gesture of /s/ is not coordinated with the preceding vowel, but is timed synchronously with /s/'s primary gesture and thus also follows the vowel. If both the TT and glottal gestures of /s/ were in

The tongue tip gesture of /s/ is also coordinated sequentially with the tongue tip gesture of /t/, and the latter is coordinated synchronously with /a₂/.

The gestural score (26) is derived from the linguistic gestural representation. The tiers show which articulator implements each gesture, and boxes show the time period during which each gesture is active (boxes at the same vertical timepoint indicate gestural overlap). This gestural score is the same as the one in (22), with vowels added. The gestural score is used to generate articulatory trajectories, overlaid in (27). The tongue tip forms a narrow constriction for /s/ and a full closure for /t/. The tongue body gesture for the two /a/ vowels is active throughout. The glottal opening gesture starts with /s/ and continues just past the release of /t/, leading to a short release. The acoustic level in (27) illustrates the approximate effect of the articulatory trajectories and alignment, but is not part of the representation.

Walker and Proctor's (2019: 470) analysis is framed in Optimality Theory (Prince and Smolensky 1993). The phonological grammar optimizes at the level of linguistic gestural representations (my (25)) (Smith 2018; Walker and Proctor 2019). Inputs and outputs are linguistic gestural representations, and constraints select the optimal one. Linear ordering is present and meaningful in these representations. Because constraints evaluate only linguistic gestural representations, candidates can be differentiated by factors like their abstract timing relations, or the presence/absence of gestures. Constraints can affect whether gestures are active at the same time (synchronous vs. simultaneous), but not their actual duration (calculated outside this phonological optimization). Even though the outputs of phonological optimization also contain the derived gestural scores and articulatory trajectories, only the linguistic gestural representations are actually evaluated by the constraints.

The constraints needed for Sevillian are as follows:

- (28) SYNC-CV: Assign * for a primary consonantal gesture that is not synchronous (in-phase) with the primary gesture of the nearest vowel.
- (29) COORD-C: Assign * for a primary consonantal gesture that is not in a coordination relation with a gesture of another segment.
- (30) WEIGHT-BY-POSITION (WBP): Assign * for a gesture that is in a sequencing relation with the preceding vowel but is not moraic (as defined by Walker and Proctor 2019).
- (31) WEIGHT-TO-STRESS PRINCIPLE (WSP): Assign * to an unstressed heavy syllable. (Prince 1990)
- (32) LINEARITY (LIN): Assign * for a pair of gestures in a sequential coordination relation whose precedence relation in the output is different than their precedence relation in the input (Walker and Proctor's 2019 LINEARITY-SEQUENTIAL).

sequential coordination relations with the preceding vowel, both would receive moras. The resulting bimoraic coda and trimoraic syllable are not desired outcomes; see Walker and Proctor (2019:477 and fn.7) for further discussion.

- (33) MAX-GESTURE: Assign * for a gesture that is present in the input but has no output correspondent.

In the tableau in (34), the input has the gestures for four segments. The order of the segments (and the gestures that constitute them) in the input is meaningful, specified in the order shown. Next to each gesture, the segment it corresponds to is in parentheses for clarity. The candidates consist of gestures, for which only constriction location is shown. /s/ is the only segment with an explicit mora; I assume vowels are inherently moraic. I do not show all possible candidates or all constraints that conflict with the ones above, and I do not attempt to establish ranking arguments between constraints.

- (34) Tableau for Walker & Proctor-style analysis of Sevillian: /a₁sta₂/

	{TB: PharNarrow ₁ } (a ₁); {TT: Critical} (s); {Glott: Open} (s); {TT: Closure} (t); {TB: PharNarrow ₂ } (a ₂)	SYNC- CV	COORD- C	WBP	WSP	LIN	MAX (GEST)
a.	$\acute{a}_1\text{-TB} \dashrightarrow \text{s-TT}_\mu \dashrightarrow \text{t-TT}$ s-Glott a ₂ -TB						
b.	$\acute{a}_1\text{-TB} \dashrightarrow \text{t-TT}_\mu \dashrightarrow \text{s-TT}$ s-Glott a ₂ -TB					*!	
c.	$\acute{a}_1\text{-TB} \dashrightarrow \text{s-Glott}_\mu \dashrightarrow \text{t-TT}$ a ₂ -TB						*!

In the tableau in (34), all candidates shown have stress, and have it correctly assigned to the penult. Because this analysis is parallel, stress assignment can be implemented in the same way as in a more standard parallel OT analysis (Section 5.1.4). In this tableau, stress is assigned but *metathesis does not occur*. I focus on preventing metathesis.

Candidate (a) (no metathesis) wins. /a₁/ is timed sequentially with the tongue tip gesture of /s/, satisfying COORD-C. This relation ensures that /s/ is moraic, satisfying WBP. The TT gesture of /s/ is coordinated sequentially with the TT gesture of /t/, and the TT gesture of /t/ is, in turn, coordinated synchronously with /a₂/ (satisfying COORD-C and SYNC-CV, respectively). Candidate (b) (metathesis) is eliminated by its LINEARITY violation (t > s instead of s > t), and candidate (c) is eliminated by its MAX(GEST) violation because it deletes the TT gesture of /s/.

Thus, stress is assigned during this level of phonological optimization (in the coupling graph), but metathesis is prevented. Metathesis would occur at other levels of representation—specifically, the gestural scores and articulatory trajectories that are derived based on the

coupling graph. The realignment that would result in metathesis was illustrated in (24). Because metathesis occurs on representations derived from the output of phonological optimization, stress assigned during this optimization is blind to the results of metathesis.

The apparently ‘opaque’ interaction between stress and metathesis falls out from the fact that stress and metathesis occur at different grammatical levels, even though phonological optimization is monostratal. By separating the two processes, this analysis accounts for the invisibility of metathesis to stress. By having metathesis operate only over gestures, this account could lend itself well to describing the variable and gradient phonetic nature of metathesis, while shielding other phonological processes from the results of gestural shifts. However, whether leaving metathesis to implementation is a desired consequence is an open question, and is discussed further in Section 5.3.

5.3 Serial vs. representational separation

Representational analyses differ substantially from process interaction analysis in how they account for the stress-metathesis interaction. The account styles differ (1) in what they assume about representations, (2) whether the processes apply in the same grammatical component vs. whether they are separated into different components, and (3) in how metathesis is driven. Analyses of both styles use ‘levels’, but these levels are qualitatively different. Representational accounts treat stress and metathesis as operating in separate, non-interacting components of the phonological grammar. In Walker and Proctor’s (2019) system, the levels consist of different layers of representation for each candidate. Although each candidate contains multiple representational layers, (non-serial) phonological optimization evaluates only one of them. This is the level at which stress is determined. Metathesis occurs in a different grammatical layer, during the implementation of gestural scores and trajectories that result from phonological optimization. In contrast, process interaction analyses treat stress and metathesis as processes that interact in the same component of the phonological grammar. In SPE and HS, the ‘levels’ represent sequential derivational levels within a single grammatical component, and both operate on the same kinds of representations. In a rule-based analysis, serialism is not necessary: both sequential and simultaneous rule application derive the same (correct) result.

Whether stress and metathesis should be controlled by the same grammar or controlled by different parts of the grammar is an open question, and one that has several consequences. For example, an HS analysis and a representational analysis like Walker and Proctor’s (2019) allow different kinds of factors to affect metathesis. In HS, metathesis can be driven by phonological constraints (e.g., to improve syllable structure). In a Walker and Proctor (2019)-style analysis, metathesis cannot be driven by similar phonological constraints, because it is separate from phonological optimization. The motivations would be limited to factors at the level of the gestural score and its implementation. On the other hand, this has a potential benefit: if the constraints evaluating syllable structure are no longer active when metathesis occurs, the [Ch] onset clusters created are less problematic.

One benefit to moving metathesis outside phonological optimization is that implementational factors can affect how metathesis occurs, and the degree to which it occurs, without changes to the output of phonological optimization. For example, the same optimal coupling graph could give rise to different degrees of metathesis at different speech rates, a result that

has been reported by Parrell (2012). Speech planning could also affect the degree of metathesis across word boundaries (since both words would need to be planned in the same chunk in order for metathesis to occur), even when the abstract coupling graphs are the same. Finally, the segment following /s/ can affect the precise articulatory implementation and acoustic result of metathesis due to its place and manner of articulation. In other words, while the abstract coupling graphs might specify the same timing relations in an /st/ and /sk/ sequence, these timing relations may show up differently on the surface due to inherent differences in /t/ and /k/.

However, leaving metathesis entirely up to implementational factors is also insufficient, because it is difficult to systematize metathesis at all. The implementation component of this model would need to be powerful in order to derive the gestural configuration that consistently results in acoustic metathesis, but also allows extensive variation and explains cross-dialectal differences in preferred variants of /sC/. On one hand, metathesis is the most frequent production of these sequences in Sevillian, and must be the preferred alignment over [hC]. On the other hand, the realization of /sC/ sequences is highly variable (Section 2.1) and metathesis applies both variably (application vs. non-application) and gradiently (degree of application). Given that /s/ still precedes /t/ in the output of phonological optimization, how does the analysis capture both systematicity and variability? Furthermore, something more is needed to explain why /sC/ sequences are produced differently across Spanish dialects (e.g., [st, ht, t, th]), if the phonological optimization outputs the same coupling graphs.

A promising avenue for a more comprehensive analysis combines a model like Walker and Proctor's (2019) with an additional grammatical component, similar to what Zsiga (2000) proposes. Following phonological optimization, the output is passed to a 'phonetic' component (terminology from Zsiga 2000) which has a different set of constraints and operates over gestural representations. It is in this component that the precise articulatory alignment of gestural targets is chosen. Zsiga (2000) also suggests that phase windows (Byrd 1996) could be used to control timing. Phase windows define a range in which gestures can be timed. The exact timing within the window is variable and influenced by language-internal and external factors. Because the framework separates the range of timing possibilities from the influences that affect timing within the range, it both allows and constrains variability. Applied to Sevillian, these phase windows would define a window of possible timing for the laryngeal and closure gestures (that may be shifting over time as metathesis develops), and also allow language-internal and external factors to affect the precise timing.

Many formal frameworks treat metathesis as transposition of two discrete segments (e.g., SPE, Chomsky and Halle 1968; OT, McCarthy 1995 and Prince and Smolensky 1993, McCarthy and Prince 1995), but the idea that metathesis may occur by manipulating gestures is not entirely new. Cross-linguistically, multiple descriptions and analyses propose explicitly, or imply, that some types of metathesis occur at the gestural level and through gestural realignment (Grammont 1933; Blevins and Garrett 1998: 510–11; Hall 2003: 4; Yanagawa 2003; Park 2006; Gilbert 2022; Gilbert and Mooney 2022; Mooney 2022).

Furthermore, treating metathesis as operating over gestures has parallels with proposals for other processes. For example, Hall (2006) proposed that some inserted vowels arise due to changes in gestural timing (intrusive vowels), whereas others are present in the abstract representation (epenthetic vowels). Each category of inserted vowel shares properties that seem to correspond with how they arise. Sevillian metathesis shares properties with Hall's

intrusive vowels. First, there is some evidence (phonetic, or reports of allophonic variants) of gestural overlap in metathesized /sC/ sequences (Alvar 1955; Hualde 1989a; Romero 1995b; Ruch 2008; Martinez-Gil 2012; Gilbert 2022). Furthermore, metathesis applies regardless of morpheme and word boundaries (Ruch 2008; Horn 2013), is not visible to stress (Ruch 2008; Torreira 2012; Horn 2013; results of current experiment), and appears to be sensitive to speech rate (Parrell 2012). A gestural account of Sevillian metathesis also provides a mechanism for how it could have developed over time through gradual shifts (Ruch 2008). The extent to which Sevillian and other metatheses can be analyzed as gesture-based, and as occurring in a separate grammatical component from processes like stress, remains to be established.

6 Conclusion

This paper investigates the interaction between stress and metathesis in Sevillian Spanish, using a restriction against the combination of antepenultimate stress and heavy penultimate syllables. The stress judgment task tests how Sevillian listeners evaluate nonce words that have antepenultimate stress and penults of different shapes—including penults followed by a stop-h sequence. If stop-h sequences derive from /sC/ clusters, then the syllable preceding the stop-h sequence is plausibly heavy at a non-surface level of representation, but light on the surface (CV.'CV.CVS.CV → CV.'CV.CV.ChV). Results suggest that participants treated penultimate syllables followed by stop-h sequences as heavy for the purposes of stress assignment, evaluating stress based on structure that is not visible on the surface. This result also supports the claim that stop-h sequences are realizations of underlying /sC/ clusters, not of a new aspirated stop category.

The mismatch between the abstract and surface forms can be accounted for in analyses of several styles. On one hand, stress and metathesis can be treated as phonological processes that interact in the same component of the grammar. Under this conceptualization, the interaction is one of countershifting opacity where one process modifies the context to which the other applies (serial), or the two apply simultaneously and are blind to each others' outputs (simultaneous). On the other hand, the 'interaction' between stress and metathesis could be conceptualized as a separation of stress and metathesis into different components of the phonological grammar, so that they operate on different levels of representation. In sum, analyses that can account for the interaction do so by making the result of metathesis invisible to stress through serialism, simultaneous application of the rules, or by separating the processes into different components of the phonological grammar. A comprehensive analysis will need to account for extensive—but constrained—variability in the application and degree of metathesis, and using gestural representations seems a promising way to do so.

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